Scalar–Tensor Hydrodynamics

Gravity as the Response of a Vacuum Medium:
A Modified Gravity Framework for Vacuum Viscosity and Cyclic Vacuum Cosmology

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Revised Formalism — Version 3.10

Abstract

We propose a hydrodynamic extension of General Relativity in which the vacuum is modeled as a dynamic, high-energy medium described by a complex scalar field $\Phi = \rho\, e^{i\theta}$, whose state is written ϕ. Rather than treating the vacuum as a fixed geometric background plus a constant cosmological term, the framework assigns the vacuum its own stress-energy, pressure, and dissipative response. The formulation separates the Einstein geometric tensor from the vacuum viscous stress tensor, introduces a dynamical scalar-field sector, replaces a fixed cosmological constant with a field-dependent pressure contribution, and allows the effective gravitational coupling to depend on the local vacuum state. The framework is intended to investigate whether cosmological expansion, galactic dynamical anomalies, and the transition between localized and delocalized excitations can arise from one underlying medium. The medium is taken as primary. What are conventionally called particles are treated as localized excitations of it, and spacetime geometry, with Einstein's equation, as the effective description of how its state is experienced. Gravity is proposed to be the medium's response to concentrated energy. The medium has one speed, at which every disturbance travels, and gravity is where that speed varies from place to place; no speed is fixed, and the measured speed of light is the local carrying speed read by rulers and clocks made of the same medium. Two principles of the framework then give General Relativity's geometry outside a non-spinning mass, including its point of no return and the observed size of a black hole's shadow.

This document is a theoretical proposal, not an established physical theory. The equations below define a research program whose consistency, stability, conservation laws, and observational predictions must be demonstrated.

Preface: What This Paper Is For

The premise

There is no empty space.

Everything this paper describes happens inside a single continuous medium, called here the Ocean. The name is a metaphor, and one feature of real oceans must be discarded at once: this Ocean has no edges. It has no shore and no surface. There is no sky above it, no floor beneath it and no air over it. It is not held in a basin, and it does not sit inside a space that could exist without it. There is no outside. The edge of what we can see is a limit of seeing, not an edge of the Ocean. Where things seem empty, the Ocean is there, quiet. Where they seem full, the Ocean is there, gathered. Nothing in this framework exists apart from it, beside it or beyond it.

What we call matter is the Ocean's own energy, gathered and circulating in place. What physics calls a particle is not an object but a pattern the Ocean keeps re-making. When a particle "moves", the place where the pattern is re-made has shifted, the way a wave crosses a stadium while nobody leaves their seat. Light is a ripple in the same medium. A galaxy is a vast region of gathered energy and current. Every pattern exists against, alongside or within others, adding to or taking from them, and every one is ultimately the Ocean's response to what the rest of it has done. For that reason this document uses the word "particle" only when reporting other authors' theories, and marks it as theirs.

The Ocean has one speed: the speed at which it carries a disturbance. Light, the inner workings of matter and every signal run at it; only the Ocean's own flow is free of it. What physics calls the speed of light is the Ocean's local carrying speed, measured everywhere with the same value because rulers and clocks are themselves patterns of the Ocean and run at the same speed. There is no fixed speed of light anywhere in this framework: a fixed number for it is a conversion between units of length and time, not a property of nature.

This work began with gravity. Einstein's theory describes what gravity does with extraordinary precision: clocks run slow near a mass, space stretches, paths curve. It does not say why concentrated energy should do this. The proposal here is that gravity is the medium's response. Concentrated energy changes the state of the Ocean around it, the Ocean runs slower there, and every pattern in it, light included, leans toward where it runs slower. On this view Einstein's equations remain the correct description, and the Ocean is offered as the reason they are correct, as the gas laws stayed correct when atoms were found to explain them. Whether the framework can carry that claim is the test it must pass: every pattern must lean by exactly the same amount, and the stretching of space must match the slowing of time exactly as Einstein requires.

The problem

The same premise bears on the largest open problem in cosmology.

Two of the largest numbers in cosmology are placeholders. Roughly a quarter of the contents of the observable universe is labeled *dark matter*: a substance inferred entirely from its gravitational pull on things we can see, and never once detected directly. Most of the remainder is labeled *dark energy*, which is less a substance than a name for an observation — that cosmic expansion is speeding up when everything we understand says it should be slowing down. Together these two labels cover something like ninety-five per cent of what the observable universe contains. Neither has an agreed physical identity.

Standard cosmology handles this competently. It adds an invisible cold matter component, and a constant term to Einstein's equations, and the resulting fit to observation is superb — good enough that the model deserves the confidence it has. But a superb fit obtained by inserting two unexplained quantities is not the same thing as an explanation. It is a description with two holes in it, and the holes have names.

This paper asks whether both holes might have one filling.

A note on where these numbers came from

There is a precedent in this subject for inserting a quantity because the answer is expected rather than because anything has been measured. In 1917 Einstein added a term to his field equations — the cosmological constant, Λ — for no reason beyond his conviction that the universe, as he conceived it, ought to be static, and the inconvenient fact that his equations refused to make it so. No observation supported the term. It was a number chosen to produce an anticipated result. When Hubble demonstrated a little over a decade later that the observable universe is expanding after all, Einstein abandoned it.

It is worth being precise about what happened next, because the common retelling is wrong in a way that matters. Λ returned in 1998, but not as Einstein's unmeasured assumption quietly surviving. It returned because observations of distant supernovae showed cosmic expansion to be accelerating, and Λ was the simplest term capable of describing that. The acceleration is real and measured. What Λ does not do is explain it: no theory predicts the constant's value, and the various attempts to calculate it from first principles fail by a margin discussed below. The term records what the expansion does and is silent on why.

Dark matter has an entirely separate and considerably stronger pedigree. Fritz Zwicky noticed in 1933 that galaxies in the Coma cluster move too fast for the visible mass to hold them together. Vera Rubin's rotation-curve measurements in the 1970s found the same discrepancy inside individual galaxies. Gravitational lensing, the structure of the cosmic microwave background, and the separation of mass from gas in colliding clusters have since confirmed it from independent directions. The evidence is not thin, and this document does not dispute any of it.

The objection is therefore not that these quantities were invented. It is that in both cases a measured discrepancy was given a name, and the name was then promoted to a substance. Λ is a parameter fitted to the expansion history rather than derived from anything. Dark matter is a material known solely by its gravity, which forty years of increasingly sensitive detectors have failed to find by any other means. Each is a placeholder that has been in service long enough to be mistaken for an answer.

If the framework set out here is even approximately correct, nothing observational changes. The rotation curves remain exactly as measured. So do the lensing maps, the microwave background, and the supernova data establishing acceleration. What dissolves is the pair of substances invoked to account for them: they stop being two unknown constituents of the observable universe and become two behaviors of one medium. The measurements survive intact. Only their cause is reassigned.

Where our two best theories disagree

General Relativity and quantum field theory are the most thoroughly confirmed descriptions of nature ever constructed, and they are known to be mutually incompatible. In ordinary practice this causes little trouble, because the two rarely describe the same situation: one governs the very large, the other the very small, and the regimes where both matter at once — the first instants after the big bang, the interior of a black hole — are exactly the regimes we cannot observe.

There is one glaring exception, and it is directly relevant here. The two theories overlap on the question of what the vacuum is, and there they do not disagree by a little. General Relativity treats empty space as genuinely empty: a smooth, featureless backdrop with no properties of its own. Quantum field theory describes the same emptiness as the most active object in physics — fields in their lowest state, never still, producing effects that have been measured in the laboratory for seventy years.

Ask both theories what that activity should weigh, and the answers differ by something between sixty and a hundred and twenty orders of magnitude, depending on where the calculation is cut off. This is not a discrepancy to be tidied away with a correction factor. It is routinely described as the worst quantitative prediction in the history of science, and it has stood unresolved for decades.

The observation that motivates this document is that the sharpest disagreement between our two best theories is a disagreement about whether the vacuum is a thing. One says no. The other says emphatically yes, and has the laboratory results to support it. What follows takes the second answer seriously and asks a further question that neither theory addresses: if the vacuum is a thing, does it also have mechanical properties — density, pressure, flow, resistance to being stirred — and if so, what would we expect to see?

What a medium can do

General Relativity, read as it is usually taught, treats the vacuum as the absence of things: a stage that can be curved by what sits on it but has no density, pressure or flow of its own. A medium does things a stage cannot. It flows. It drags. It carries waves. It can be stirred, and it pushes back when you stir it. Each of those behaviors is a candidate for a phenomenon currently attributed to something unknown:

- A galaxy rotating inside a medium that may swirl along with it would experience stress from that swirl; whether the medium does swirl with it is one of the things the framework must still derive. Stress gravitates. The extra pull that standard cosmology assigns to a halo of undetected particles, in its own terms, might instead be the vacuum's own response to being rotated.
- A medium with internal pressure that can change character over time can drive expansion without a constant pasted into the equations by hand. What is now called dark energy would be the Ocean in a particular state rather than a number in a table.
- And — this is the part developed furthest here — a medium has a smallest meaningful size. Below some scale, it cannot be disturbed in a localized way at all. That gives a principled distinction between what counts as a localized pattern and what counts as a ripple.

The ambition is that what are called dark matter and dark energy stop being two mysteries and become two behaviors of one thing.

Black holes

A black hole, on this account, is the Ocean's largest whirlpool. It does not gather a substance; it gathers organization. Whatever falls in is stripped of its own pattern until it turns with the whirlpool. Black holes recycle; they do not drain. Most of what they return comes back early, as light and jets from the matter swirling around them, and the rest leaves over immense times as faint radiation. Energy returns to the Ocean; structure does not. Nothing is gathered at its center: what falls in flows through and returns to the Ocean, much later. Its horizon is a point of no return, not a surface and not an edge of the Ocean, which stays whole across it. Section 8.2 follows a single traveler in.

What this document is not

It is not a theory. It is a proposal with its homework listed.

The equations below are consistent in form, but consistency in form is cheap. A physical theory has to survive conservation laws, stability analysis, the recovery of tested results in the appropriate limits, and quantitative agreement with observations that were not used to build it. None of that has been done here. Section 12 is an itemized list of the work required, and several items on it could terminate the framework outright. They are stated plainly rather than buried, because a proposal that cannot specify how it would fail is not worth the paper it occupies.

How this document came to exist

The framework began as private speculation, written to a standard that required only that the pieces appear to fit. That is a much weaker standard than the one it has since been held to, and the gap between the two turned out to be where the interesting work was.

Seven revisions have followed, and they share a shape.

The first corrected a structural error. In the original formulation the vacuum's viscous response *replaced* the Einstein tensor — the geometric heart of General Relativity. This cannot work: a mathematical identity guarantees that the Einstein tensor is conserved, and substituting something else for it breaks the conservation of energy and momentum. The fix was to keep the geometry intact and add the vacuum's stress alongside it rather than in its place.

The second removed a number. The original drew the line between matter and vacuum at $0.511\,\mathrm{MeV}$ — the rest energy of the electron. Above the line, matter; below it, vacuum. This was an acknowledged placeholder, and it had three problems: the value was imported rather than derived, it was observer-dependent (two observers moving relative to one another would disagree about which side of the line a given thing fell), and it was an abrupt cutoff where physics prefers continuity.

The replacement drops the number entirely and asks instead how small a disturbance the medium can hold. Every medium has such a scale — in a laboratory superfluid it is called the healing length, the shortest distance over which the medium can bend without tearing. Disturbances larger than it spread out and remain part of the medium. Disturbances smaller than it stay put, as localized patterns. The matter/vacuum distinction becomes a property of the Ocean rather than a line drawn across it.

The result worth noting is what happened next. Calibrating that scale to present conditions puts the Ocean's healing length at roughly 386 femtometers. (A femtometer is a millionth of a billionth of a meter, about the size of a proton.) That is exactly the electron's own characteristic length. Every pattern has a natural spread set by its mass, tighter the heavier it is, and the Ocean can hold in place only a pattern whose spread fits within its healing length. The electron's spread is 386 femtometers, so the electron is simply the lightest pattern the Ocean can hold in place, and everything lighter, photons and neutrinos included, spreads too wide and is necessarily a ripple instead. The healing length is the Ocean's tightest grip, not the electron's size: probed directly, the electron shows no size at all down to a scale several

hundred thousand times smaller. The number that had been an unexplained input came back out as a consequence. That is the kind of exchange that makes a framework worth continuing.

The third responded to news from outside. In September 2026 a proof was announced that a fluid with internal friction can, under smooth external stirring, tear itself into a singularity in finite time — a result about ordinary fluid dynamics with no connection to this framework whatsoever. It remains unverified and does not transfer directly to a relativistic medium. But it removed an assumption this framework had been resting on without noticing: that viscosity is enough to keep a continuum well-behaved forever.

The repair was already present. The smallest-scale limit introduced in the second revision is precisely what stops a disturbance concentrating without bound — but only if the medium resists compression rather than welcoming it. That condition had never been written down. It is now a stated requirement, and if the Ocean turns out to have the other sign, the framework fails as written.

The fourth was a change of mind. The first revision had kept Einstein's geometry intact and set the Ocean beside it, and every version since had carried that arrangement forward: a stage, and a medium placed upon it. The stage was never argued for. It was inherited, from the habit of treating spacetime as the thing that exists and fields as things that happen in it. Asked directly what the stage was made of, the framework had no answer, and a medium that fills everything leaves no room for a second structure to occupy.

So the stage is removed. There is only the Ocean. Einstein's equation stays, because it describes what is measured with extraordinary precision, but its standing changes: it is now the description of how the medium's state is experienced by patterns made of that medium, not a separate structure the medium sits in. Gravity, on this account, is the medium's response to concentrated energy. That claim brings obligations the earlier arrangement did not carry, and they are set out in the technical sections; the heaviest is to derive the geometry rather than assume it.

The fifth removed a picture. Earlier versions borrowed the image of a black hole as a drain, with the medium pouring inward. Taken literally the image fails: a medium flowing in would have to pile up somewhere, and nothing piles up. Near a black hole the Ocean's state changes steeply, and that change bends every path until, at the horizon, nothing can travel outward; the medium itself stays where it is. The same reasoning gave gravity and expansion one root. Both are a change in the Ocean's state: where the change varies from place to place it bends paths, and where it is the same everywhere at once it makes the Ocean flow apart.

The sixth found the stage still standing in the equations. The fourth revision had removed it as a position, but the law written for the Ocean still measured every disturbance against a fixed speed that belonged to no part of the Ocean. Light took its speed from the Ocean's state; matter took its limit from that fixed speed; and no arrangement of the two could produce gravity. So the second speed is removed. The Ocean has one speed, the speed at which it carries a disturbance. Light, the workings of matter and every signal run at it; only the Ocean's own flow is free of it. Gravity is where that speed varies from place to place.

The seventh is the one this version records, and it found the stage once more, this time in the arithmetic. The sixth had removed the second speed in words, but every equation for the Ocean was still written in the standard way, with a fixed speed of light built into how a change in time is weighed against a change across space. So the fixed speed is removed. There is no constant c anywhere in the law: every speed in it is the Ocean's own carrying speed where it is, and the value measured everywhere is how that speed reads to rulers and clocks made of the same Ocean.

Each revision has the same form: something the framework had not earned is removed, whether it was chosen by hand, imported or inherited, and what replaces it must then be derived rather than assumed. That a framework can absorb that process repeatedly is not evidence that it is true. It is evidence that it is the right kind of thing to keep testing — which is a lower bar, and the only one this document claims to clear.

A note on method, since it is unusual enough to be worth recording. The revisions above were worked out in dialogue with a large language model, used as an interlocutor rather than an author: a way of being argued with at the speed of thinking.

What would kill it

Five things, named here so that a reader does not have to hunt for them.

The scale gap. For the Ocean to explain galactic rotation it must be able to act across thousands of light years. For it to place the matter boundary at the electron it must have structure at the scale of a hundredth of an atom. These two requirements differ by about thirty-two orders of magnitude. They are set by different parts of the mathematics, so this is not an outright contradiction — but a framework that merely tolerates such a gap has not explained anything, and this is the most serious unresolved problem in the document.

A route out now exists, and it is recorded as a route rather than a result. A hurricane is hundreds of miles across, but that size is not a property of air. It belongs to the storm: to how much energy and spin were gathered in one place. Air itself has only small-scale properties. If the same holds here, the galactic scale is not built into the Ocean at all. A galaxy is not an object that stirs the medium; it is a region of the medium carrying excess energy and excess current, of which the visible stars are the dense middle and the wide slow flow is the outskirts. The Ocean would then need only one built-in size, the small one, together with the ability to carry very long, slow disturbances at almost no cost. Laboratory superfluids have that ability for a known reason, a symmetry of the underlying field. The price is that the field must be a richer object than a single real number at each point, which the framework now adopts (Section 2.2), and that the calculation showing such a disturbance reproduces observed galaxies has not been done. Until it is, the gap stays on this list.

The route does not make the gap smaller. A galaxy really is about 10^{32} times larger than the Ocean's smallest structure, just as a hurricane is about 10^{14} times wider than the spacing of air molecules. What the route changes is whether that ratio needs explaining. If the large scale belongs to the disturbance and not to the medium, it is the size of a storm, not a second setting of the air.

The sign. As above: if the Ocean's self-interaction has the wrong sign in its present state, nothing arrests collapse and the framework is not viable in its present form. This is a single calculation away from being settled either way. Settling it for the Ocean's present state does not settle it for every time and place. The medium resists compression only in some of its possible states; in others it may not, and the earliest moments of the present beat or distant regions could have been in such a state. The rules do not change from place to place, but where the medium sits within them can.

The preferred frame. Describing the vacuum as a medium gives it a state of rest, and a state of rest is something Einstein spent a career removing from physics. Real superfluids do exactly this, so it is not absurd — but the experimental limits on any such effect are extraordinarily tight, and the framework must show it stays inside them. Among the tightest are the comparisons of matter with antimatter, which agree to parts in a trillion (Section 10, deficiency (5)).

The measured vacuum. Empty space has already been measured doing things. It screens electric charge, it scatters light off light, and in very strong magnetic fields it splits light by polarization, as a crystal does. Standard physics predicts these effects by other means, in some cases to many decimal places. If the Ocean is real, these are its properties, and it must reproduce the same numbers. Agreement would be strong support. Any disagreement would be fatal.

The strength of gravity. Gravity is extraordinarily weak. Between two electrons the electric force is about 4×10^{42} times stronger than their gravity; that ratio is measured, not assumed. If gravity is the medium's response, the medium must produce it. In the best-known scheme for deriving gravity from a medium the strength is set by the medium's shortest length, and with the Ocean's that would make gravity vastly too strong. Either the Ocean has structure at a much smaller scale, which it must then explain, or its response sets the strength of gravity some other way. Expressed as a length, the gap is the distance from the Ocean's smallest structure down to the Planck length, about 10^{22} times smaller; but the Planck length is built from the strength of gravity and has never been measured, so the measured ratio is the honest statement. The medium sits between two gaps it must account for: the weakness of gravity below its own scale, about 10^{22} when written as a length, and the size of galaxies, about 10^{32} above.

To which should be added a possibility that no calculation will resolve: the agreement between the Ocean's smallest scale and the size of the electron may simply be a coincidence. The Standard Model

already accounts for the electron's mass by other means. Whether these are two descriptions of one fact or two unrelated facts that happen to agree is, at present, unknown.

What follows is the technical statement of all of this. It is written for a reader comfortable with tensor notation, but the argument it makes is the one above: that the emptiness between things may not be empty, and that a great deal of what we currently call unknown may be the behavior of a single medium that has been sitting in plain view the whole time.

Conventions. No independent particulate entities are posited. An excitation of the medium that is localized in the sense of Section 7 is called a *localized excitation*; one that is not is a *coherent excitation*. "Matter" and "the matter sector" denote the localized population of the one medium, never a second substance. Particle-like behavior — the exchange of energy, momentum and charge in whole units at a single point — is treated as a property of how excitations of a quantum field interact, not as evidence of objects (Section 7). The word "particle" appears only when describing other work, and is marked as that work's term. The framework is four-dimensional throughout: indices μ, ν run over one dimension of time and three of space, and the Ocean is a medium in all four. Time is part of the medium's description, not a clock standing outside it.

In the paper's own voice "motion" is not used without a qualifier: it is always flow, circulation or travel, as defined below. Every measurement is a measurement of travel, since instruments are themselves patterns; flow is inferred from its effects on travel, never observed directly. The medium is written as the complex field $\Phi = \rho\, e^{i\theta}$ with density $n \equiv \rho^2$ (Section 2.2); to keep θ for the phase, the expansion scalar of the medium's flow is written $\Theta \equiv \nabla_\alpha u^\alpha$. In the paper's own voice c means the local carrying speed and c_0 its measured value. Wherever a speed only converts between units, as in GM/c_0^2 or $\hbar c_0$, it is written c_0: the number a local clock and ruler return, not a fixed speed of nature. "Information" appears only when reporting the physics literature, as in the black-hole information problem; in the paper's own voice the term is *state*. "Dark matter" and "dark energy" appear only as what other work calls them, never as components of this framework: the observations they name stand, and their interpretation here is the coherent extent of a galaxy's disturbance (Section 6) and the Ocean's own pressure $P(\phi)$ (Section 5). The hottest stage of a beat transition is *the hot, dense phase* or *the transition plasma*; "primeval fireball" is the literature's term. Neither is a ball: it has no surface, center or outside, and is the condition of the whole Ocean at once.

Glossary. The following terms are used with fixed meanings throughout.

The Ocean
The single continuous medium. It has no edges and no outside, and it includes every beat. Nothing in the framework exists apart from it.

Flow
The Ocean's own motion, described by the medium four-velocity u^μ. Flow is not limited by the carrying speed c_s; where it outruns c_s for an observer, there is a horizon (Section 11.1).

Carrying speed
c_s, the speed at which the Ocean carries a disturbance: the one speed of light, of the internal workings of matter and of every signal. It has two readings. On the spot it is always measured with the same value; seen from afar it varies, and that variation is gravity (Section 11.1).

Speed of light
c; not a constant of nature in this framework. The Ocean's local carrying speed, as read by rulers and clocks made of the same Ocean; c_0 is the value so measured, $299\,792\,458\ \mathrm{m\,s^{-1}}$, the same everywhere because the instruments scale with the speed they measure. No fixed c appears anywhere in the Ocean's law; a fixed number for it is a conversion between units of length and time (Section 11.1).

Circulation
The Ocean's motion within a pattern. A pattern can circulate internally while staying in place.

Travel
A pattern's motion as a whole: the location at which the Ocean re-makes it, shifting. Travel is measured by the group velocity of the excitation relative to u^μ, and is limited by the local carrying speed.

State
The complete configuration of the Ocean's flow and circulation at the finest scale. What physics calls information.

Pattern
The form of an excitation: organized structure recognizable at some scale, such as the shape of a whirlpool or the profile of a wave. Patterns can be scrambled; the state is not destroyed.

Carried quantities
What a pattern carries: energy, momentum, circulation and charge. Two patterns of the same form can carry very different amounts. The conserved carried quantities are what survives annihilation, Hawking radiation and a beat transition.

Excitation
The Ocean's departure from its background state at some location: a pattern together with what it carries. Matter and light are excitations; so is a galaxy.

Light
A free ripple of the Ocean, all of its carrying speed spent in travel. Matter is a confined ripple, its carrying speed spent circulating in place: same medium, same speed, different shape.

Healing length
ξ, the shortest distance over which the Ocean can bend before it springs back; about 386 femtometers in the present state (ξ_0), set by the amount ρ. It is the threshold below which an excitation localizes, not the size of any excitation and not a grain of the medium.

Localized excitation
An excitation localized in the sense of Section 7; what other work calls a particle.

Coherent excitation
An excitation that is not localized.

Circulation sense
The property distinguishing a localized excitation from its antimatter partner. It is not spin and is not changed by any rotation (Section 7.7, item (8)).

Horizon
A point of no return for travel: before it, escape is possible; after it, it is not. Not a physical thing and not a surface, and never a boundary of the Ocean, which remains continuous and coherent across it. Horizons are observer-dependent; a region beyond ours has its own horizon, and ours lies within it.

Matter, matter sector
The localized population of the Ocean, never a second substance.

The Heartbeat
The whole sequence of beats.

A beat
One expansion and contraction of the Ocean, from one beat transition to the next.

The present beat
The beat we are in, and the only one any observation can reach.

Beat transition
The change of state between one beat and the next. Never a phase change within a beat.

The big bang
The most recent beat transition, as seen from within the present beat. Not the beginning of the Ocean.

Epoch
A phase within a beat, such as the radiation-dominated, matter-dominated or present accelerating epoch. Never a beat.

The present expansion epoch
The hot expanding phase we observe, about 13.8 billion years old by extrapolation. It contains the

radiation-dominated, matter-dominated and present accelerating epochs. Distinct from the present beat, whose age is not known (Section 5.2).

The observable universe

The part of the present beat that observation can reach. Neither the whole present beat nor the Ocean.

"Universe" is not used on its own, since what it names is redefined here; it appears only in "the observable universe" and in reporting other work. "Cycle", like "particle", appears only when reporting other work. "Cyclic" in the subtitle names the established category of cosmology to which the Heartbeat belongs.

1 The Field Equation

The governing equation for the "Ocean" metric is formulated as Einstein geometry coupled to a dynamic scalar vacuum medium. The central revision relative to the original formulation is that the viscous response is treated as a *stress contribution* rather than as a replacement for the geometric Einstein tensor:

$$G_{\mu\nu} + P(\phi)\, g_{\mu\nu} \;=\; \kappa(\phi)\Big[\, T^{\text{matter}}_{\mu\nu} + T^{\phi}_{\mu\nu} + \Pi_{\mu\nu}\,\Big], \qquad \kappa(\phi) = \frac{8\pi G(\phi)}{c(\phi)^4}. \tag{1}$$

Here $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}R\, g_{\mu\nu}$ remains the geometric Einstein tensor; $P(\phi)$ is the dynamic vacuum pressure; $T^{\phi}_{\mu\nu}$ is the stress-energy of the scalar field; and $\Pi_{\mu\nu}$ is its hydrodynamic dissipative stress.

Status of the equation. Equation (1) is retained as the working equation of the framework, but its standing has changed. Earlier versions treated $G_{\mu\nu}$ as a fundamental structure with the medium added to it. Here the medium is primary: the geometry on the left-hand side is the effective description of how the medium's state is experienced by excitations of the medium, and the right-hand side records the medium's own stresses. The equation is kept because Einstein's description is correct to the precision it has been tested; the framework's claim is to explain why it holds (Section 2.5), and deriving it from the dynamics of the medium is the central task that claim creates (Section 12).

2 Tensor Definitions

2.1 Geometric Einstein tensor

The geometric sector retains the standard covariant structure of General Relativity:

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}R\, g_{\mu\nu}. \tag{2}$$

This separation is essential because the contracted Bianchi identity gives $\nabla^{\mu} G_{\mu\nu} = 0$. Any additional vacuum stresses must therefore satisfy the corresponding total conservation requirement (Section 9).

2.2 Scalar vacuum stress-energy $T^{\phi}_{\mu\nu}$

The vacuum is represented by a dynamical scalar field rather than a purely phenomenological density. Let

$$X \equiv -\tfrac{1}{2}\,\nabla_{\alpha}\phi\,\nabla^{\alpha}\phi, \tag{3}$$

and let $L(\phi, X)$ be the scalar-field Lagrangian. Its stress-energy tensor is

$$T^{\phi}_{\mu\nu} \;=\; L_{,X}\,\nabla_{\mu}\phi\,\nabla_{\nu}\phi \;+\; L\, g_{\mu\nu}. \tag{4}$$

This term gives the Ocean a genuine dynamical energy density and pressure. The specific choice of L determines whether the field behaves as a conventional scalar, a k-essence-like medium, or a more specialized superfluid effective theory. Section 7 depends on this choice directly.

The complex field. The Ocean is represented as a complex field with a global $U(1)$ symmetry,

$$\Phi = \rho\, e^{i\theta}, \qquad n \equiv \rho^2, \qquad \theta \;\to\; \theta + \text{const.} \tag{5}$$

This is the baseline representation, and it may be the bosonic sector of a richer order parameter (Section 7.7, item (7)). Four independent requirements select it over a single real scalar: the separation of density, flow and current below; antimatter partners (Section 10, deficiency (5)); a carried circulation sense (Section 7.7, items (8)–(9)); and the gapless phase on which the candidate route for the scale hierarchy depends (Section 7.7, item (5)).

Where expressions in this paper are written for a single real field ϕ, ϕ stands for the state of the medium, which the modulus carries: $\phi \leftrightarrow \rho$. The state-dependent quantities $P(\phi)$, $\kappa(\phi)$, $\xi(\phi)$, $\eta(\phi)$ and $\zeta(\phi)$ are functions of ρ and of the phase rate μ defined below, the quartic potential of Section 9.1 is a potential for ρ, and the beat transitions of the Heartbeat are transitions of ρ. The shift symmetry forbids any dependence on the value of θ itself, so the phase has no potential: its long-wavelength excitation is a gapless Goldstone phonon, as in a laboratory superfluid, and the mass-gap entry of the table in Section 7 vanishes for it. The excitation of the modulus is heavy: it is the amount mode, whose minimum energy is the cost of a dent, and its quanta are the localized excitations of Section 7.

The two components play different parts in the Ocean's state. The amount ρ sits at the minimum v of its potential, held there by the potential rather than by any conserved quantity, so it does not thin as the Ocean expands; it departs from v only in extreme states, at a beat transition or in the core of a collapsed object. The second state variable is the rate at which the phase turns, the Ocean's chemical potential

$$\mu \equiv \sqrt{-\nabla_\alpha\theta\,\nabla^\alpha\theta}, \qquad u_\mu = -\frac{\nabla_\mu\theta}{\mu}, \tag{6}$$

which sets the Ocean's pressure and defines its rest frame: the medium four-velocity u^μ lies along the gradient of the phase. The $U(1)$ symmetry gives a conserved charge, with density proportional to $\rho^2\mu$. As the Ocean expands that charge spreads and μ falls, while ρ stays at v. In a beat, μ is the clock and ρ is what resets at the transition. A medium with $\mu = 0$ would have no rest frame at all.

The density n measures how much of the Ocean is present, the gradient of the phase θ sets the flow velocity, and their product is the current: how much of the Ocean is moving. A pattern is the form these quantities take; its carried quantities are integrals over them. A single real field has no phase, so this separation is not available to it. Nothing within a pattern is carried along as a bulk substance; the current, not a moving volume, is the measure of how much of the Ocean is in flow.

The choice brings the framework's field content closer to the complex-field description of Huang and collaborators [1] and to the phonon description of Berezhiani and Khoury [2]; the distinctive claim of Section 10 is unaffected. The complex field does not supply a crossing of $w = -1$ (Section 5.1).

In the present state the Ocean is stiff: its amount ρ barely gives, and the ratio $2\mu^2/(\lambda v^2)$ is below about 10^{-23} (Section 11.1). Seen from afar, its phase turns at one rate everywhere, so the phase is one clock for the whole Ocean; on the spot it turns faster wherever clocks run slow (Section 11.6).

2.3 Vacuum viscosity tensor $\Pi_{\mu\nu}$

The viscous response represents the dissipative stress response of the scalar medium to local flow. For a relativistic effective fluid, a first-order constitutive form is

$$\Pi_{\mu\nu} = -2\,\eta(\phi)\,\sigma_{\mu\nu} \;-\; \zeta(\phi)\,\Theta\,h_{\mu\nu}, \tag{7}$$

where $\eta(\phi)$ is the shear viscosity, $\zeta(\phi)$ the bulk viscosity, $\Theta = \nabla_\alpha u^\alpha$ the expansion scalar, $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$ the spatial projector orthogonal to the medium four-velocity u^μ, and $\sigma_{\mu\nu}$ the shear tensor.

In regions with coherent vorticity or differential flow, $\Pi_{\mu\nu}$ can generate additional gravitational stress. The model therefore investigates whether the vacuum response can reproduce effects conventionally assigned to dark-matter halos. The term *rotational support* is used here as a hypothesis about the gravitational effect of the vacuum stress response; viscosity itself is not assumed to behave as additional localized mass.

An effective description. $\Pi_{\mu\nu}$ is dissipative and therefore irreversible; taken as fundamental, it would destroy fine-grained state everywhere, contrary to Section 9.2. It is read instead as the coarse-grained stress of the quantized vortices of Section 7.7, item (5) — their Reynolds and mutual-friction stress, averaged over scales well above the intervortex spacing — with the underlying dynamics conservative. It therefore has no meaning below that averaging scale, and mutual friction requires a normal component that the formalism has not yet defined (Section 12). A normal component would also bring a second sound with a speed of its own, which the one-speed principle does not allow to carry signals that patterns can use (Section 11.1); how the two fit is open.

2.4 Scalar pressure $P(\phi)$

$P(\phi)$ replaces a fixed cosmological constant as the phenomenological vacuum-pressure contribution. It is a function of the state of the underlying scalar field rather than a universal constant:

$$\Lambda g_{\mu\nu} \longrightarrow P(\phi)\, g_{\mu\nu}. \tag{8}$$

The *Heartbeat* hypothesis proposes that the scalar field can evolve through alternating dynamical phases. An effective positive-pressure / negative-pressure transition may then produce expansion ("diastole") and contraction ("systole") phases. A complete model must derive this behavior from a field potential or equation of state rather than prescribing an oscillating pressure by hand.

2.5 Effective coupling $\kappa(\phi)$

The coupling between the vacuum state and ordinary stress-energy is allowed to depend on the local scalar density:

$$\kappa(\phi) = \frac{8\pi G(\phi)}{c(\phi)^4}. \tag{9}$$

$G(\phi)$ represents a field-dependent effective gravitational coupling. The quantity $c(\phi)$ is the Ocean's local carrying speed, the one speed at which every disturbance travels (Section 11.1). Locally it is always measured as c_0, because rods and clocks are patterns of the Ocean and scale with it. Seen from afar it varies, and that variation is gravity (Section 11.1).

The refractive reading. The account of gravity adopted here is refractive. A wave in a medium whose propagation speed varies curves toward the region where it travels more slowly. If concentrated energy lowers the local propagation speed of the Ocean, light curves inward, and so does every localized excitation, each of which is a pattern the medium keeps re-making. Where the medium is slower on one side, the re-making is uneven and the pattern is re-made slightly further toward the slow side. Nothing pulls; trajectories are turned. This is consistent with the absence of any felt force in free fall. In the weak field it reproduces the standard equivalence between gravitational light bending and a refractive index

$$n(r) \simeq 1 + \frac{2GM}{rc_0^2}. \tag{10}$$

Refraction alone does not reproduce the full weak-field and post-Newtonian behavior — perihelion advance, frame dragging, gravitational radiation — and extending it to do so is the test this reading must pass (Section 12). Three requirements are decisive. Every excitation must respond to the medium's state in exactly the same proportion, since all bodies and light are observed to fall alike to extraordinary precision; here that equality must follow from everything being made of the same medium. The response must slow time and stretch space in equal measure: the measured deflection of starlight by the Sun is twice what slowing alone would give, and the second half comes from the spatial part. And disturbances of the response itself, gravitational waves, must travel at the speed of light, as observed when a neutron-star merger's gravitational and light signals arrived together. Section 11 states the second requirement as a condition on how patterns scale with the carrying speed (Section 11.5) and offers a candidate mechanism

for it (Section 11.6). The reading of $c(\phi)$ as a genuine second causal speed, kept as a fallback in earlier versions, is set aside: there is one speed (Section 11.1).

3 Scalar-Field Dynamics

The field ϕ must have an independent equation of motion. A generic covariant form is

$$\Box\phi \;=\; \frac{\partial V_{\text{eff}}}{\partial \phi} \;+\; C(\phi, T, \nabla\phi, \ldots)\,, \tag{11}$$

where $V_{\text{eff}}(\phi)$ is an effective potential and C represents permitted couplings to matter, curvature, or hydrodynamic variables. The potential and coupling structure determine the vacuum phases, stability, characteristic sound speed, and possible Heartbeat behavior.

4 The Ocean Interpretation

The framework interprets the observable universe as a dynamical region of a deeper vacuum medium. The Ocean is not required to begin at the big bang. Instead, the big bang is interpreted as a beat transition: a change of state of the medium, a phase transition, or a rapidly expanding solution of the underlying field equations.

The observable universe is therefore not treated as the ontological boundary of reality. It is the portion of the deeper continuous vacuum field accessible to observation. This distinction is a conceptual premise of the framework and must ultimately be connected to a mathematically well-defined global solution.

5 Cosmological Heartbeat

The proposed *Heartbeat* cosmology associates cosmic expansion and contraction with the evolution of ϕ. A schematic beat is

ϕ-state transition $\rightarrow$ pressure reversal $\rightarrow$ expansion $\rightarrow$ field evolution $\rightarrow$ contraction $\rightarrow$ new phase.

The model does not require that each beat create the underlying vacuum field. Instead, each *beat* (see the Glossary) may be a dynamical state of the same deeper medium.

Two kinds of statement are kept apart in what follows. Section 5.1 concerns the present beat, which is the only one any observation can reach. Section 5.2 concerns the sequence of beats and the transitions between them, about which no observation made within a beat can speak directly.

5.1 Within the present beat

Everything known about time is known from within the present beat. Every observer and every instrument is a pattern of the Ocean in this beat, and every observation is of this kind. Here time is the pace at which the Ocean re-makes its patterns, measured by clocks that are themselves such patterns. Durations are meaningful and measurable, but local: clocks run at different rates in different places, as General Relativity describes and as the account of Section 2.5 requires.

Expansion is divergence, not transport. Expansion is an effect of flow whose cause is a change of the Ocean's state. On the largest scales the Ocean's flow is divergent, $\Theta = \nabla_\alpha u^\alpha > 0$ everywhere, with no center and no common direction. A uniform flow carrying every pattern the same way would be unobservable in principle; what is observed is regions moving apart. Regions beyond the Hubble radius recede faster than light. This is flow, not travel, and no pattern exceeds the local speed limit. Some galaxies now receding faster than light remain visible, because the Hubble radius has grown to overtake light they emitted long ago [3]. Bound systems decouple from the divergence: galaxies and planetary systems do not stretch internally.

One root, two effects. Both gravity and expansion begin with a change of the Ocean's state, and differ in its shape. A gradient of state, varying from place to place as near a mass, bends travel: it refracts patterns and, where steep enough, brings outward travel to zero at a horizon, with no net flow of the medium (Section 8.1). A uniform change of state, the same everywhere at once, drives flow: a divergent flow with no center, which is expansion, or a convergent one, which is contraction. In the terms of Section 2.2 the uniform change is a change of μ, with the amount ρ held at v. Literal flow is excluded for a black hole because a local inflow would have to accumulate somewhere; a uniform flow has no somewhere, and the change of state is the same everywhere at once. Nor does literal flow conflict with a constant vacuum energy density: a medium with negative pressure does work on itself as it flows apart, adding energy at the rate that keeps its energy density fixed. The cosmic horizon is therefore a flow horizon and a black-hole horizon a gradient horizon, two mechanisms with one root. Deriving this principle from the dynamics is a task of Section 12.

The pace of the beat. The pace of the beat is not fixed. While matter dominated, its gravity slowed the expansion; distant supernovae show that the expansion is now accelerating [4, 5]. There are further indications that the acceleration is itself evolving: baryon acoustic oscillation measurements from the DESI survey, combined with the microwave background and supernova samples, prefer what the survey calls a dark-energy component that changes over time to a constant one, at roughly 3 to 4σ depending on the supernova sample, short of the level conventionally required for discovery [6]. In this framework such variation is the Ocean's state changing through $P(\phi)$, not a departure from a constant. One caution follows from the same data. The preferred evolution crosses the boundary $w = -1$, which a single scalar field with a standard kinetic term cannot readily do; the survey's authors note that multiple fields, additional internal degrees of freedom or a non-minimal coupling can evade the difficulty [6]. The complex field of Section 2.2 does not supply this: it is two canonical fields, and canonical fields cannot cross $w = -1$. The crossing needs a phantom component or a non-minimal coupling. If the evolution is confirmed, the Ocean's pressure sector would need such structure, and the candidate already present is the coupling $\kappa(\phi)$.

5.2 Across beats

Nothing in this subsection is observable from within the present beat. It states what the framework permits, not what has been found, and the two subsections should not be read into one another.

No clock survives the transition between beats. The patterns that keep time are diluted or remade, and the Ocean's state, including ξ, may differ from one beat to the next. Durations therefore cannot be compared from beat to beat, and "faster" or "slower" across beats can only mean that different beats pass through different sequences of states. The framework requires no repetition: each beat may differ in its expansion, its reset and its ξ. Across beats the Heartbeat is counted, not timed. Nor is a beat set to a length: it is event-driven, ending when its triggering condition is met, not when a span of time has passed. If that condition is that no differences remain to drive any flow, a black hole still returning what it holds is unfinished, and the end of the beat waits for it; most if not all black holes are gone before a beat ends, and any that remain are taken up by the transition. The return times of Section 11.8 then set a lower bound on the length of a beat.

Whether such a sequence can avoid singularities, entropy accumulation and instabilities is an open mathematical question. Since the state is never destroyed (Section 9.2), whatever resets disorder at a beat transition acts at the level of patterns, not by erasing the state. Stiffness keeps the Ocean regular against cascades (Section 9.1) but cannot turn a contraction around: for any material whose pressure does not exceed its energy density, contraction once under way can only speed up (Section 11.8). **[Candidate]** What turns a black hole's collapse around, a loosening of the Ocean's grip under extreme compression, is the candidate for what turns a beat's contraction around: one mechanism for both.

The nearest established precedent is the cyclic model of Steinhardt and Turok [7], in which a scalar field rolling in a potential drives an endless sequence of expansions and contractions, each big bang being

a transition from a preceding contraction with finite temperature and density. Two of its answers bear directly on the questions above. Entropy does not accumulate from cycle to cycle because a long phase of accelerated expansion, the present accelerating epoch, dilutes the matter, radiation and black holes of the previous cycle before contraction begins. And the nearly scale-invariant density fluctuations are generated during the slow contraction rather than by inflation, with the distinguishing prediction of no long-wavelength primordial gravitational waves. Both are available to the Heartbeat in principle. The difference lies at the bounce: in that model the transition is motivated by colliding branes along an extra spatial dimension, with the four-dimensional field theory carrying a matching rule supplied by hand. The premise of this framework admits no space outside the Ocean, so the bounce must instead come from the medium itself, through the healing-length floor and the state dependence of its stiffness (Section 9.1).

One consequence of this route can be tested from within the present beat. If the Heartbeat seeds the density fluctuations of the present beat during the preceding slow contraction, as in the Steinhardt–Turok model, rather than by inflation, it predicts essentially no long-wavelength primordial gravitational waves. The present upper limit on their amplitude relative to the density fluctuations, from Planck combined with BICEP2/Keck Array data, is $r < 0.056$ [8], and forthcoming microwave-background experiments will test it much further. A detection of primordial gravitational waves would count against a contraction-seeded Heartbeat; continued non-detection would leave it viable. This route also avoids a difficulty noted for inflation in this framework: the field values that inflation requires lie far above the scale set by ξ_0.

Two obligations follow from setting inflation aside. *The horizon problem*: the microwave background has the same temperature, to about one part in 10^5, across regions that have not been in contact within the present expansion; the framework's account is that the previous beat's contraction brought them into contact before the transition, and it must be shown that this produces the observed uniformity and its spectrum of departures. The same account must also supply flatness, which inflation otherwise explains: in the cyclic model the long accelerated expansion of each cycle flattens and smooths what follows [7], and the Heartbeat would need its pressure-driven expansion to do the same. *The age of the Ocean*: the 13.8 billion years is the age of the present expansion epoch, extrapolated on the assumption that patterns have kept the same time throughout, not the age of the beat or of the Ocean. Whether the Ocean has an age at all turns on whether disorder accumulates from beat to beat and whether the Heartbeat has no net expansion from one beat to the next; both are open (Section 12).

6 Galactic Dynamics

In the galactic regime, the hypothesis is that spatial gradients, vorticity, and coherent flow of the vacuum field generate an additional effective stress. The observable circular velocity would then be determined by the combined baryonic and vacuum-field gravitational response:

$$v_c^2(r) \;=\; v_b^2(r) \;+\; v_{\phi,\mathrm{eff}}^2(r). \tag{12}$$

Here v_b is the baryonic contribution and $v_{\phi,\mathrm{eff}}$ denotes the effective contribution produced by the scalar field and its stress response. The goal is not to insert an arbitrary halo profile, but to derive $v_{\phi,\mathrm{eff}}(r)$ from ϕ, $\Pi_{\mu\nu}$, and the field equations.

Rotation is a special case. The relation above applies to rotationally supported disks. Galaxies also carry random travel of their stars, inflow and outflow of gas, vertical oscillation, bars and spiral patterns, and travel through their surroundings. Pressure-supported systems with little net rotation — elliptical galaxies and dwarf spheroidals — follow the same radial acceleration relation as disks [9]. Whatever links the vacuum contribution to the baryons therefore cannot depend on rotation, and the framework must apply to pressure-supported galaxies as well.

One structure, not two. The decomposition above is bookkeeping and should not be read as a statement that a galaxy acts on the medium from outside. In this framework a galaxy is a region of the Ocean

carrying excess localized energy and elevated current. Its baryonic content is the population of localized excitations of that region and its extended coherent excitation is the population of phase ripples, in the sense of Section 7; the two are the dense interior and the diffuse exterior of a single solution of the field equations. This differs from treatments in which galaxies are external sources perturbing the superfluid [1].

Two consequences follow. First, the kiloparsec scale is a property of the individual solution, fixed by the full state of the disturbance — its energy, its organized and random flow, and its exchange with the surrounding medium — and need not be a parameter of the medium; this is the basis of the candidate route of Section 7.7, item (5). Second, if the solution is fixed by its baryonic content — the question posed by the Size task of Section 12 — the baryonic distribution determines $v_{\phi,\mathrm{eff}}(r)$ with no freedom adjusted galaxy by galaxy. The observed radial acceleration relation has that form: across rotationally supported galaxies the total acceleration is fixed by the baryonic acceleration alone, with little or no intrinsic scatter and no sign of a second parameter [10, 9]. That is a target to be derived, not a result.

It also sharpens the exposure recorded in Section 10. A contribution fixed by the baryons is precisely the kind that risks being parallel to the baryonic gravitational field, and a parallel contribution is excluded by Milky Way vertical kinematics [11, 12]. The single-structure reading therefore needs a vacuum contribution whose magnitude is set by the baryons while its direction is not slaved to theirs. Whether the field equations permit that is not known.

The vacuum contribution is a flow. The decomposition $v_c^2 = v_b^2 + v_{\phi,\mathrm{eff}}^2$ reduces the vacuum's response to a single radial quantity. In the framework it is a flow in time and space together, not a single radial number. Momentum conservation means that any local acceleration of the medium produces shear against neighboring regions; the shear term $\sigma_{\mu\nu}$ of $\Pi_{\mu\nu}$ is where this enters. Such a flow is not constrained to lie parallel to the baryonic gravitational field.

The route proposed here is shear bound to the galaxy itself: the differential circulation of a flattened vortex lattice in the baryons' gravitational well. Azimuthal and meridional shear of such a lattice gives an anisotropic $\Pi_{\mu\nu}$, and it does not depend on the galaxy's travel through its surroundings. That independence is required: the radial acceleration relation shows almost no intrinsic scatter across galaxies traveling at very different speeds relative to the cosmic rest frame, and in colliding clusters the inferred extra mass follows the galaxies rather than the slowed gas. A wake from a galaxy's travel, aligned with its direction of travel rather than with the baryonic field, remains only as a small residual offset, tested separately (Section 12). Stress in the Ocean does not push stars; it reaches them only through how patterns read the Ocean's state (Section 11.3). Whether the lattice's anisotropy reaches the stars as the radial-over-vertical enhancement the parallelism criterion requires is therefore the calculation.

A swirl that pulls only inward. **[Candidate]** One form does it. In the slow limit of the Ocean's rulers and clocks with the Ocean flowing at $\mathbf{v}$, a pattern moves as

$$\ddot{\mathbf{x}} = -\nabla\Phi_b + (\mathbf{v}\cdot\nabla)\mathbf{v} - \mathbf{w}\times(\nabla\times\mathbf{v}), \qquad \mathbf{w} = \dot{\mathbf{x}} - \mathbf{v}, \tag{13}$$

with Φ_b the baryons' potential and $\mathbf{w}$ the pattern's travel relative to the Ocean; that patterns read the flow in this form is assumed, not yet derived. For a flat swirl of the Ocean around the galaxy, $\mathbf{v} = V\,\hat{\boldsymbol{\phi}}$ with V the same at every radius over the range considered and at every height near the disk, a star on a circular orbit at speed v_c feels an extra acceleration $v_c V/R$, entirely inward; a star at rest relative to the galaxy feels none, and vertical travel is untouched. If V changes with radius, the inward pull gains a term $(v_c - V)\,dV/dR$. In a model Milky Way whose stars and gas alone give $159\,\mathrm{km\,s^{-1}}$ at 8.1 kpc against the observed 229, a swirl of about $118\,\mathrm{km\,s^{-1}}$ supplies the rest, and a star's vertical oscillation matches the baryons-only case to within two per cent: extra radial pull with none vertically, the criterion superfluid dark matter fails [12]. The swirl must not weaken with height near the disk: if it falls off over half a kiloparsec, the same star is thrown ten kiloparsecs out of the disk. A superfluid supplies exactly this shape, since it rotates only through vortex lines, straight and parallel to the axis, so the swirl they carry is the same at every height. The vortex spacing implied, with the condensing mass at m_e, is about

1200 km, the same order as the spacing of Section 7.7, item (5). The extra pull depends on how a star travels through the swirl, which Gaia's velocity data can test; light crossing a galaxy should be deflected differently on the side moving with the swirl and on the side moving against it, which lensing must be checked against; and what sets the swirl's strength from each galaxy's baryons, as the radial acceleration relation requires, is not yet given.

Stars and gas orbiting a galactic center are travel. Whether the Ocean also flows with them is a separate claim, about how the travel of patterns couples to the flow of the medium, and the formalism does not yet specify that coupling. The decomposition $v_c^2 = v_b^2 + v_{\phi,\mathrm{eff}}^2$ treats the baryons as travel and the vacuum contribution as flow; the relation between the two must be derived, not assumed.

7 The Localization Threshold

7.1 Why a fixed threshold fails

The earlier formulation partitioned energy between sectors by a declared inequality:

$$E > 0.511\,\mathrm{MeV} \;\Rightarrow\; \text{localized matter sector}, \qquad E \le 0.511\,\mathrm{MeV} \;\Rightarrow\; \text{scalar-fluid density } \phi.$$

This has three defects, only the first of which is usually noticed.

(1) **It is an unexplained parameter.** The value is imported from the electron rest mass, which the framework does not derive.

(2) **It is not covariant.** E is frame-dependent. A boost changes which sector a given excitation belongs to, so the partition is observer-dependent — unacceptable in a theory built on the covariant field equation (1).

(3) **It is discontinuous.** A step function requires an exchange term coupling $T^{\mathrm{matter}}_{\mu\nu}$ and $T^{\phi}_{\mu\nu}$. Combined with the variable coupling $\kappa(\phi)$, this makes the conservation conditions of Section 9 essentially impossible to satisfy.

The revision removes all three by replacing the declared threshold with a property of the vacuum field's own excitation spectrum.

7.2 The excitation spectrum of ϕ

Perturb the field about a background configuration,

$$\phi(x) = \bar{\phi}(x) + \delta\phi(x), \tag{14}$$

and let u^μ be the four-velocity of the vacuum medium with spatial projector $h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$. Expanding the Lagrangian $L(\phi, X)$ of Section 2.2 to second order in $\delta\phi$ gives a quadratic action of the form

$$S_2 = \frac{1}{2}\int \sqrt{-g}\left[\mathscr{K}\left(u^\mu \partial_\mu \delta\phi\right)^2 - L_{,X}\, h^{\mu\nu}\partial_\mu\delta\phi\,\partial_\nu\delta\phi - V''_{\mathrm{eff}}\,\delta\phi^2 - \mathscr{G}\left(h^{\mu\nu}\partial_\mu\partial_\nu\delta\phi\right)^2\right], \tag{15}$$

with $\mathscr{K} \equiv L_{,X} + 2XL_{,XX}$. The final term is a gradient-stiffness (quantum-pressure) operator, which earlier versions imported from the non-relativistic superfluid and used to set the threshold. With the law written on the Ocean's own rulers and clocks (Section 11.1) the Ocean is stiff and that term carries no signal at any accessible energy (Section 7.7, item (3)); the threshold rests instead on the gap.

Varying (15) and evaluating in the medium rest frame, the Ocean has two branches. The phase ripple has no gap and travels at the carrying speed, $\omega = c_s k$. The amount mode has a gap:

$$\boxed{\omega^2(k,\phi) \;=\; \Omega^2_{\mathrm{gap}}(\phi) \;+\; c_s^2(\phi)\,k^2} \tag{16}$$

Its minimum energy, $\hbar\Omega_{\mathrm{gap}}$, is the cost of a dent.

7.3 The three characteristic scales

Every coefficient in (16) is fixed by the Lagrangian rather than chosen; the gap belongs to the amount mode, and the phase's gap vanishes (Section 2.2):

Quantity	Definition	Physical meaning
Sound speed	$c_s^2 = L_{,X}/\mathscr{K}$	Carrying speed of the Ocean, the speed of every disturbance (Section 11.1)
Mass gap	$\Omega_{\rm gap}^2 = V_{\rm eff}''(\bar{\phi})/\mathscr{K}$	Curvature of the effective potential at the local vacuum state
Mass of the amount mode	$m_\star = \hbar\Omega_{\rm gap}/c_s^2$	The cost of a dent, $m_\star c_s^2$, expressed as a mass

From these follows the single scale that carries the whole section — the *healing length*:

$$\xi(\phi) \;=\; \frac{\hbar}{m_\star(\phi)\, c_s(\phi)}. \tag{17}$$

ξ is the reduced Compton length of the amount mode: the shortest distance over which the Ocean's amount can be bent before the cost of a dent dominates. It is the Ocean's localization scale: a threshold, not a granularity (Section 7.7, item (7)).

7.4 The localization criterion

The partition is by branch. Quanta of the amount mode are localized excitations and contribute to $T^{\rm matter}_{\mu\nu}$; phase ripples are coherent excitations and contribute to $T^{\phi}_{\mu\nu}$. Earlier versions partitioned a single branch by wavelength, placing localized excitations on the k^4 part of the spectrum; that part carries no signal in the stiff Ocean, and the partition moves to the branches. The dimensionless quantity

$$\mathscr{Q}(\phi,k) \;\equiv\; \xi^2(\phi)\, h^{\mu\nu} k_\mu k_\nu, \tag{18}$$

constructed covariantly from an excitation's four-momentum and the medium's own state, now says how a quantum of the amount mode travels: for $\mathscr{Q} \ll 1$ it is slow, $\hbar\omega \approx m_\star c_s^2 + \hbar^2 k^2/2m_\star$, a pattern at rest or traveling slowly; for $\mathscr{Q} \gg 1$ it travels near the carrying speed, $\omega \approx c_s k$. Both are matter. The threshold energy is the cost of a dent itself,

$$E_c(\phi) = \hbar\Omega_{\rm gap}(\phi) = m_\star(\phi)\, c_s^2(\phi), \tag{19}$$

the rest energy of an amount quantum, the same for every observer.

The partition is by branch, not switched. $T^{\rm matter}_{\mu\nu}$ is the energy–momentum of the amount mode's quanta and $T^{\phi}_{\mu\nu}$ that of the phase ripples. Since both are populations of the *same* field, their sum is conserved identically, and exchange between them is the field's own interaction, as when a pair of amount quanta is made from energy of at least twice the cost of a dent. No exchange term is required, and the consistency conditions of Section 9 are no longer obstructed by the threshold.

7.5 Recovering 0.511 MeV as a calibration

Requiring $E_c(\phi_0) \approx 0.511\,\text{MeV}$ in the present vacuum state fixes

$$\xi(\phi_0) = \frac{\hbar c_0}{E_c} \approx 386 \text{ femtometers}, \tag{20}$$

which is the reduced Compton wavelength of the electron. This is a derived consequence rather than a coincidence. A localized excitation of rest energy mc_0^2 has Compton wavelength $\bar{\lambda}_C = \hbar/mc_0$; it can localize within the medium only if $\bar{\lambda}_C \lesssim \xi_0$, which requires

$$m \gtrsim \frac{\hbar}{c_0\,\xi_0} = m_e. \tag{21}$$

The electron is therefore the lightest excitation the present vacuum can hold as a localized pattern. Anything lighter has a Compton wavelength longer than the Ocean's localization scale and cannot be localized in it — it is necessarily a coherent field excitation. This is offered as the framework's explanation for why photons and neutrinos behave as field phenomena while the electron is the floor of the localized sector. The value 0.511 MeV is demoted from axiom to measurement: it is the readout of $\xi(\phi_0)$ for the present beat, not an input. A candidate reading, not yet a result: the minimum energy of a ripple in the amount ρ, the cost of disturbing how much Ocean is present, is the natural candidate for E_c. If it holds, 0.511 MeV is the price of a dent in the Ocean, and the electron-scale coincidence becomes a statement about the medium (Section 12).

7.6 Consequences

The threshold is local and dynamical. $E_c = E_c(\phi(x))$, so regions in different vacuum states have different matter thresholds. This connects the present section to the Heartbeat cosmology: a phase transition of ϕ is also a transition in what the vacuum is *able to hold* as matter. Where ξ grows, the localization floor rises and previously localized states are no longer supported. Within a beat such departures are confined to extreme states, such as a beat transition or the core of a collapsed object; elsewhere the threshold holds the value fixed for the beat, as the next consequence states.

The electron's mass is a constant of the beat. The value 0.511 MeV, with $\xi_0 \approx 386$ femtometers, remains a readout, but of the present beat rather than of the present moment: wherever the Ocean is in its ordinary state it is fixed for the whole beat by the condition in which the Ocean was left at the transition, and another beat may read differently. The scale is set by the amount ρ, which its potential holds at v; the change of state that drives the flow apart (Section 5.1) is a change of μ, and leaves ρ, and with it the electron's scale, steady throughout the beat. The medium is not conserved as a quantity; its state is governed. This agrees with the limits on drift of the proton-to-electron mass ratio, parts per million over billions of years from quasar absorption spectra [13] and a few parts in 10^{17} per year from atomic clocks [14]. Two consequences follow. The conserved charge of the $U(1)$ symmetry is no threat to this: its density is proportional to $\rho^2\mu$, and it dilutes with expansion through μ, not through ρ (Section 2.2). And the question is no longer how fast m_e drifts but what sets the beat's constant at the transition. Under the one-speed principle (Section 11.1) a uniform change of the carrying speed, across a beat or within one, would be invisible on its own, since every pattern scales with it; what these limits test is whether any pattern fails to scale exactly.

So is the strength of gravity. $G(\phi)$ is a readout of the same held state, and so is the Planck length assembled from it, so the ratio $\alpha_G = (\ell_P/\xi_0)^2$ of (23) is fixed for the beat as well (Section 7.7, item (5)). Lunar laser ranging over fifty years finds $\dot{G}/G = (-5.0 \pm 9.6) \times 10^{-15}$ per year [15]; if that rate has been steady, G has changed by less than about one part in 10^4 over the present expansion epoch.

It introduces a preferred frame. The dispersion relation (16) is stated in the rest frame of u^μ. This is honest — real superfluids do exactly this — but it is a departure from exact local Lorentz invariance and must be reconciled with precision tests.

Particle-like behavior does not require smallness. An excitation appears particle-like because it exchanges energy, momentum and charge in whole units and at a single point. Both properties follow from the medium being a quantum field with local couplings, independently of the localization scale. The healing length is a threshold for localization, not a radius, and above it the term used here is *excitation* or *coherent excitation*, whose size is set by circumstance.

7.7 Open items specific to this section

(1) **The amount mode.** Earlier versions imported a gradient-stiffness term from the non-relativistic Bogoliubov form to define $m_\star$ and ξ. On the Ocean's own rulers and clocks that term carries no signal, and ξ is the reduced Compton length of the amount mode, set by its gap. The gap must be derived from the Lagrangian; matching it to 0.511 MeV is a calibration (Section 12, "The cost of a dent").

(2) **Stability.** Require $L_{,X} > 0$ and $\mathscr{K} = L_{,X} + 2XL_{,XX} > 0$; otherwise $c_s^2 < 0$ (gradient instability) or the kinetic term flips sign (ghost).

(3) **Lorentz-violation bounds.** Under the one-speed principle (Section 11.1) c_s is the local c_0 by construction, and the test becomes whether light and matter share it; their speed limits agree to about one part in 10^{23} (Section 11.4). The k^4 term of the dispersion relation gave short-wavelength excitations of ϕ a group velocity above c_s, which the one-speed principle does not allow for anything that carries a signal. With the law written on the Ocean's own rulers and clocks the Ocean must be stiff, and in the stiff Ocean the coefficient of the term falls to about $4a^2 \lesssim 10^{-46}$ (Section 11.1), so it carries no signal at any accessible energy. Had the term been significant, it would have acted at wavelengths near ξ_0, at energies near 0.5 MeV; photons from distant gamma-ray bursts show no energy-dependent arrival times at energies far above this, which excluded it from the electromagnetic sector and from electrons. Stiffness now removes it everywhere. Relatedly, a gamma-ray photon has a wavelength far below ξ_0; under the partition by branch it is not a quantum of the amount mode and is not classified as localized. The matter–antimatter comparisons of Section 10, deficiency (5), are among the most precise of these tests.

(4) **Varying-m_e bounds.** Quasar absorption spectra constrain drift in m_p/m_e at roughly the 10^{-5}–10^{-6} level over cosmological time; the Oklo natural reactor and atomic clocks constrain it locally. Earlier versions required any Heartbeat evolution of ξ to be negligibly slow over the epochs these measurements probe and largely confined to beat transitions. That is now the position (Section 7, Consequences): ξ is a constant of the beat, and the bounds confirm it.

(5) **The scale hierarchy.** Galactic-scale effects require the scalar's Compton range to be of order a kiloparsec, implying $\Omega_{\rm gap} \sim 10^{-26}\,$eV. But $\xi_0 \approx 386$ femtometers implies a stiffness scale near 0.5 MeV. These differ by roughly thirty-two orders of magnitude. They belong to different branches — the phase, whose potential vanishes by symmetry, and the amount, whose gap is the cost of a dent — so this is not immediately inconsistent, but a credible model must *explain* the separation rather than merely tolerate it. This is the most serious open problem in the revised section.

Candidate route. The estimate $\Omega_{\rm gap} \sim 10^{-26}\,$eV assumes that the kiloparsec scale is intrinsic to the medium. On the single-structure reading of Section 6 it is instead a property of the individual galactic solution. The requirement on the medium then weakens from a tuned value to an upper bound,

$$\Omega_{\rm gap} \lesssim 10^{-26}\,\text{eV}, \tag{22}$$

which $\Omega_{\rm gap} = 0$ satisfies. A vanishing gap is protected, rather than tuned, because the long-wavelength excitation is the phase θ of the complex field, whose $U(1)$ shift symmetry (5) forbids it a potential; a small nonzero gap from soft breaking of that symmetry would be technically natural. The medium would then carry one intrinsic length scale, ξ_0, and the hierarchy would be a ratio between a parameter of the medium and the size of a particular solution, which is not a naturalness problem. In the same construction the heavy modulus is the amount mode, whose gap sets ξ_0 (item (1)).

None of this is established. The route trades the hierarchy for three obligations: to show that the symmetry survives the couplings of equation (1); to say what fixes the size of a galactic solution; and to pass the vertical-acceleration test in the form stated in Section 6. It also carries the consequences recorded in Section 2.2. Until those are met the hierarchy stands as an open problem.

The route does not remove the ratio between ξ_0 and galactic scales; it changes it from a separation between two parameters of the medium into the size of a solution relative to the medium's one intrinsic length. A consistency check follows from the circulation quantum. A superfluid carries rotation in quantized vortices whose areal density is fixed by the local rotation rate and the mass of the condensate quanta (Feynman's rule). For quanta at the mass implied by ξ_0, and the rotation of the solar neighbourhood ($220\,\mathrm{km\,s^{-1}}$ at 8 kpc), the vortex spacing is of order 600 km, so the medium is smooth on galactic scales and the route is not contradicted by granularity. The estimate takes the carrying speed at its measured value c_0 and assumes local rigid rotation. Because the medium then carries a length and no intrinsic acceleration, the acceleration scale $g_\dagger$ of the radial acceleration relation [10] must arise elsewhere, most plausibly from the cosmological background state.

The scale gap has a twin. The strength of gravity at the electron's scale is the measured dimensionless ratio

$$\alpha_G \equiv \frac{G m_e^2}{\hbar c_0} = \left(\frac{\ell_P}{\xi_0}\right)^2 \approx 1.75 \times 10^{-45}, \tag{23}$$

against $\alpha \approx 1/137$ for electromagnetism. Written as a length, it places the Planck length $\ell_P = \sqrt{\hbar G/c_0^3} \approx 1.6 \times 10^{-35}$ m about 10^{22} below ξ_0. The Planck length is not measured; it is assembled from $\hbar$, G and c_0, the present values of $G(\phi)$ and $c(\phi)$, and since those depend on the Ocean's state (Section 2.5) it is, here, a readout of that state rather than a fundamental length. If gravity is the medium's response, the medium must produce α_G; schemes that set the strength by the medium's shortest length would, with ξ_0 as that length, make it vastly too strong. Either the Ocean has structure far below ξ_0, a second scale to explain, or the refractive route sets G some other way. This is the hierarchy problem in the Ocean's form.

(6) **Relation to the Standard Model.** The Higgs mechanism already accounts for m_e via a Yukawa coupling. The framework must state whether ξ_0 and the electron Yukawa are independent quantities that happen to agree, or whether one is meant to determine the other.

(7) **Size and spin of the localized excitation.** ξ_0 is the scale below which an excitation localizes, not its radius. Quantum electrodynamics has the same separation: the electron's reduced Compton wavelength limits how far it can be localized without pair creation, while scattering and its magnetic moment show no internal structure down to about 10^{-18} m. The framework must show that its localized excitation responds as a point to probes at that resolution. A second route exists: $\xi(\phi)$ is field-dependent, so the local healing length inside a concentrated excitation may be far smaller than ξ_0. That route is constrained from the other side, since the healing length must not shrink without bound in states where the medium is stiff, or the cutoff of Section 9.1 disappears.

Separately, the electron has spin $\frac{1}{2}$. Excitations of a scalar field, real or complex, carry integer spin. The localized sector therefore requires structure that the present scalar formulation does not contain, and this applies to every fermion, including the neutrinos described above as field phenomena. A candidate direction is a multi-component order parameter, spinor or matrix-valued as in superfluid helium-3 A, which near topologically protected Fermi points supports chiral spin-$\frac{1}{2}$ quasiparticles and effective gauge fields with relativistic dispersion [16], and would also supply the second internal phase of item (8). Light's two transverse polarizations and the two tensor polarizations of gravitational waves make the same demand (Section 11.9). Its cost is foundational: in helium-3 those quasiparticles exist because the atoms beneath are fermions, so this route makes the Ocean fermionic at bottom.

(8) **Circulation sense is invariant under rotation.** In the winding reading, a localized excitation and its antimatter partner are the same pattern with opposite circulation sense. An ordinary vortex cannot carry this: viewed from the other side it has the other sense, so rotation would turn matter into antimatter, which observation forbids. The framework needs two properties, an orientation in space that rotation changes (the candidate for spin) and a circulation sense that only pair creation or

annihilation changes. Item (9) favors an internal-phase charge over a knotted pattern, and it cannot be the condensate's own phase θ, whose current is the Ocean's own current; it requires a second internal phase, and so an order parameter with more than one component (item (7)). Neither route is derived here.

(9) **The conserved sense is shared across the threshold.** Total lepton number has never been observed to change, while in beta decay an electron appears with an antineutrino, which by Section 7 is a coherent excitation, not a localized one. Whatever plays the role of lepton number must therefore be carried by coherent excitations as well as localized ones and pass between them; a winding carried only by localized cores would not do.

8 Limiting Regimes

Regime	Required behavior	Physical objective
GR / Solar System	$\phi \to \phi_0$; $G(\phi) \to G_0$; $c(\phi) \to c_0$; vacuum stresses negligible	Recover tested General Relativity
Galactic	$\Pi_{\mu\nu}$ and scalar gradients become significant	Explain rotation curves without dark matter in the particle form its proponents posit
Cosmological	$P(\phi)$ and $T^{\phi}_{\mu\nu}$ contribute on large scales	Generate the expansion history, including the acceleration other work attributes to dark energy
Beat transition / Heartbeat	ϕ evolves between dynamically distinct states; $\xi(\phi)$ shifts	Provide a mechanism for the expansion and contraction of beats and for changes in the localization floor
High-energy	Energy reaches twice the cost of a dent, so pairs of amount quanta form	Test the derived localization threshold of Section 7
Strong field / black hole	Point of no return at $2GM/c_0^2$; collapse turned around by a loosening of the Ocean's grip, not by stiffness	Match horizon-scale imaging and merger ringdowns; calculate the turnaround and the return

8.1 Black holes

In the refractive reading of Section 2.5, a black hole is a region where waves are slowed enough that an outward-directed wave is turned back. Laboratory analogs make this concrete. A fluid flowing faster than its own sound speed forms a horizon that sound cannot cross, and such a horizon has been made in an atomic Bose–Einstein condensate, where the spontaneous analog of Hawking radiation was observed together with the entanglement of its pairs [17]. In that system the shortest length scale is the healing length $\xi = \hbar/mc_s$, the same construction as Section 7. General Relativity itself admits an exact description of a stationary black hole as space flowing inward through a flat background, reaching the speed of light at the horizon [18]. For a rotating black hole that description has the river twist rather than swirl. A draining vortex in water is nonetheless an analog of a rotating black hole, and rotational superradiance — waves extracting energy from the rotation — has been observed in one [19].

The horizon is refractive. The medium does not pour radially inward. A literal inflow faster than the wave speed would, by continuity $\nabla_\mu(n\,u^\mu) = 0$, make the Ocean accumulate inside the horizon. (Near the horizon the Ocean does keep in step by a gentle inflow, Section 11.8; that is not this.) The inflow of the river description is instead an effective geometry, produced by the steep radial change in the Ocean's state and in $c(\phi)$ near the mass, and the horizon is where outward travel falls to zero. Laboratory horizons,

which are made by literal flow, therefore share the black hole's mechanism without being the same kind of system. This is the gradient half of Section 5.1.

No accumulation. A black hole is not gathered substance. On the premise that matter is energy circulating in place, a black hole is a large organized circulation, and its growth is the growth of that organization. Nothing piles up. Mass is the total energy of the circulation; the black hole's spin in the sense of General Relativity is the net circulation after many contributions on different axes cancel. A non-rotating black hole is not still; its circulations sum to zero.

Stripping is the no-hair theorem. Infalling patterns are scrambled until only the black hole's mass, net rotation and charge remain visible from outside, as the no-hair theorem requires. Frame dragging forces nearby patterns into co-rotation. Circulation sense is among what is stripped: a black hole formed from antimatter is indistinguishable from one formed from matter, so the black hole is the one setting in which the matter/antimatter distinction is undone without annihilation. The distinction between what is stripped and what survives has a standard reason. Electric charge is coupled to a long-range gauge field, so it can be measured by a surface integral far from the black hole, and survives as hair. Lepton and baryon number are global symmetries with no long-range field, and are stripped; circulation sense is stripped because it is a global internal charge (Section 7.7, item (8)). The same reasoning sets an obligation: the Ocean has no long-range gauge field yet, and electric charge requires one. Stripping scrambles patterns; it does not destroy the state (Section 9.2).

Currents near a rotating black hole. A current that runs across a pattern's path faster than the pattern can travel does not prevent its passage; it carries it sideways and prevents it from holding still. That is the ergosphere of a rotating black hole, lying outside the horizon, where nothing can remain at rest relative to distant regions but escape is still possible, and where rotational energy is extracted. A horizon, by contrast, stops travel in one direction altogether, and encloses what lies inside it.

Hawking radiation is the escape route. The black hole's energy, and with it the state, leaves as Hawking radiation. At the horizon the steep gradient of the Ocean's state, which brings outward travel to zero, stretches fluctuations of the medium into real outgoing ripples, the energy drawn from the black hole. A fluid flowing faster than its own speed of sound has a horizon of the same form, and is predicted to emit thermal sound by the same argument [20]; a laboratory analog has since observed the effect [17]; in the Ocean the gradient of state does the same work. The ripples carry energy and momentum together. For a rotating black hole they carry angular momentum several times faster than energy, so it spins down to a nearly non-rotating state before it has given up most of its mass [21]; rotational energy can also be drawn off by superradiance, the amplification of waves by a body that turns faster than the wave pattern sweeps round it [22, 19]. For a non-rotating black hole the radiation is isotropic and carries energy with no net momentum.

The temperature is set by how steeply outward travel falls to zero at the horizon: the surface gravity in General Relativity, the velocity gradient in laboratory analogs, and the gradient of the Ocean's state here. The horizon is defined by that condition, so it does not change in character as the black hole shrinks; it moves inward. A shrinking black hole therefore becomes tighter, not faster: its gradient steepens and it radiates hotter and faster, giving the runaway of the final stage. Rotation lowers the temperature, to zero at maximal spin. The radiation emerges from a region comparable in size to the black hole, its typical wavelength being about eighty times the horizon radius [23], not from a glowing surface. What an observer detects depends on their state of travel: a pattern falling freely detects nothing at the horizon, while one held stationary near it detects a thermal bath. In the Ocean's terms, what a pattern detects depends on its travel relative to the medium.

It will not be measured astrophysically. For a solar-mass black hole the Hawking temperature is about 60 nK and the mass loss about $10^{-45}\,\mathrm{kg\,s^{-1}}$, far below what it absorbs from the microwave background;

every astrophysical black hole currently gains mass. Evaporation times run from about 10^{67} years upward: Page's original calculation, which treated neutrinos as massless, gives 2×10^{66} years for a solar mass [24], and without massless neutrinos the figure is several times longer. Searches for final-stage bursts from small primordial black holes have found none and constrain their abundance. The mechanism is tested through laboratory analogs, which share the black hole's mechanism, a horizon at which outward travel stops, although their horizon is made by literal flow and the Ocean's by a gradient of state.

Black holes recycle; they do not drain. A black hole is not a passage to anywhere else; there is nowhere else. It returns energy to the Ocean and keeps no structure. Most of the return happens outside the horizon and early. Accretion converts about 6 per cent of infalling rest mass into radiation for a non-rotating black hole and up to about 40 per cent for a rapidly rotating one, against 0.7 per cent for nuclear fusion. A magnetic field threading a spinning hole can draw off its rotational energy electromagnetically and beam it out in two opposite directions [25]; the largest jets so powered reach millions of light-years, and the outflow heats and redistributes the surrounding gas, regulating star formation. The Fermi bubbles above and below the Milky Way's center were most likely created by a large release of energy at the center within the last ten million years or so, either an accretion episode of the central black hole or a nuclear starburst, the discoverers finding difficulties with both [26]. Mergers return a few per cent of the mass as gravitational waves, and Hawking radiation returns the rest. What crosses the horizon returns only as energy: no atom, electron or quark comes back out as itself. Any localized excitations in the output are made afresh by the Ocean, as electron–positron pairs from black holes below about 10^{14} kg, and hadrons in the final stage [24]. Jets contain matter, but matter from the accretion flow that never crossed. Energy is recycled; structure is not, though the state is preserved (Section 9.2).

Three questions remain.

The core. Earlier versions held that, with $\lambda > 0$, collapse might stop at a core of finite density set by the healing length. It cannot: stiffness cannot reverse collapse inside a point of no return, and the healing length's density scale is far below that of a neutron star. A loosening of the Ocean's grip can, and the flow then passes through and returns (Section 11.8). This is the question Section 5 asks of a contraction phase, posed for a single object.

Two horizons. The authors of the river model stress that its flat background has no observable meaning [18]. Earlier versions allowed a geometric horizon and a horizon of the medium that need not coincide. With the law written on the Ocean's own rulers and clocks (Section 11.1) there is one field of rulers and clocks for every pattern, so the two coincide: the geometry is the medium's own state, which bears directly on the structural question of Section 10. The wavelength-dependent leakage once allowed by the k^4 term goes with that term in the stiff Ocean (Section 7.7, item (3)).

Observation. Horizon-scale imaging and gravitational-wave ringdowns agree with General Relativity, so any departure must lie inside the horizon or below current sensitivity.

Two further obligations are recorded in Section 12. A black hole's entropy, the size of the state it can hold, is proportional to its horizon's area rather than its volume, which a medium holding state throughout its volume must explain. And a black hole traveling relative to the medium, as a recoiling merger remnant does, should show asymmetric infall and possibly drag; General Relativity does not distinguish that from being carried by the flow, but a medium with a rest frame does.

8.2 A traveler falls in

This subsection follows a single pattern, a human traveler in an ideal protective suit, into a black hole, and marks at each stage what is established, what is the framework's reading, and what is open. It is a living section, reviewed at each release, and its standings change as the items they depend on change. **[Established]** means General Relativity and observation; **[Framework]** means the Ocean's reading, consistent with what is established; **[Open]** means it depends on unresolved items.

Choice of black hole. **[Established]** Tidal stretching acts on every part of a body at once, so no suit resists it. Near a black hole of ten solar masses the head-to-foot difference in acceleration reaches about ten times Earth's gravity some 3800 km out, while the horizon lies at about 30 km: the traveler is torn apart long before arrival. The case followed here is the black hole at the center of the Milky Way, Sagittarius A*, of about 4.3 million solar masses and horizon radius about 12.7 million km, where the head-to-foot difference at the horizon is about a ten-thousandth of Earth's gravity. **[Framework]** What tears a traveler apart is not the swiftness of the Ocean's current but how steeply it changes across a body (Section 11.2). Near a stellar-mass black hole the current steepens over a few kilometers; at Sagittarius A* it is nearly uniform across a body at the horizon, and the traveler crosses intact, carried by a swift current that is barely felt.

Approach. **[Established]** Outside the horizon the danger is the accretion flow, gas at millions of degrees radiating X-rays, which the suit is assumed to withstand. Around a rotating black hole frame dragging turns the traveler with it, and inside the ergosphere no pattern can remain at rest relative to distant regions. At 1.5 horizon radii light can orbit, and that orbit sets the size of the black hole's shadow; for Sagittarius A* the observed shadow lies within about ten per cent of General Relativity's prediction [27]. **[Framework]** The traveler is a pattern traveling through a region where the Ocean's rulers and clocks are increasingly squeezed and outward travel, seen from afar, increasingly slows. Around a non-spinning mass the Ocean's geometry is General Relativity's (Section 11.7), so light orbits at the same place and the shadow has the same size. Outside the horizon outward travel is still possible, and so is escape.

The horizon. **[Established]** The traveler notices nothing at the horizon: no wall, flash or jolt, as the equivalence principle requires. Distant observers never see the crossing: the traveler's light reddens, slows and fades, and the last image dims at the edge. The traveler's own clock runs normally. **[Framework]** The horizon is a point of no return: before it, escape is possible; after it, it is not. It is not a physical thing, not a surface, and not an edge of the Ocean, which is whole and connected across it. The same point holds for every pattern, since every pattern has the same carrying speed. In the Ocean's geometry the current through time is brought to zero, seen from afar, at $2GM/c_0^2$, General Relativity's radius (Section 11.7). Near it the Ocean keeps in step with itself by flowing gently inward, and the traveler, falling with it, meets nothing there: the Ocean's own reason for the established line above (Section 11.8). Falling freely, the traveler detects no Hawking radiation. Past the horizon every direction of travel leads inward.

Inside. **[Established]** The longest possible time from horizon to center is $\pi GM/c_0^3$: about 66 seconds for Sagittarius A*, about 28 hours for M87* at 6.5 billion solar masses. Firing rockets shortens it. The outside sky remains visible, distorted; a falling traveler does not see the whole future of the observable universe pass. Tides grow during the fall and become lethal about 280 000 km from the center of Sagittarius A*, stretching bodies, then molecules, then atoms. **[Framework]** Inside, forward in time points inward, and the Ocean's own phase flows inward with the traveler. A pattern's time is its rhythm; as the traveler's patterns are stripped, their clocks go with them.

The core. **[Framework]** Nothing is gathered at the center. Recycling, for the Ocean, is flow-through: the inward flow does not stop or collect at the center, but passes through and returns to the Ocean, later. Time is recycled with it: inward on the way in, outward again on the way out. **[Candidate]** General Relativity ends the fall at a singularity, where the theory fails. The Ocean's resistance to compression ($\lambda > 0$, Section 9.1) cannot turn the inward flow around once it is inside the point of no return; a loosening of the Ocean's grip under extreme compression can (Section 11.8), stopping the fall short of a point and turning it around about half a minute after the crossing, on the traveler's own clock, for Sagittarius A*. **[Open]** The return must be slow enough, seen from afar, to fit everything observed of black holes. How slow depends on what the Ocean is made of; written as a function of that fine scale it is slow enough, but it is not yet derived. Separately, the localization floor is local (Section 7): if the core's state raises

ξ, localized excitations are no longer supported there, and the traveler's localized patterns dissolve into coherent excitations rather than being crushed. Whether compression raises or lowers ξ is not known.

What becomes of the traveler. **[Framework]** The traveler's patterns are scrambled until nothing distinguishes them from the black hole. The carried quantities, energy, momentum and charge, add to its mass, net rotation and charge. The state is preserved (Section 9.2) and returns to the Ocean. **[Open]** By which route: as Hawking radiation, over about 10^{87} years for Sagittarius A*; through the flow-through of the core, of order 10^{22} years or longer seen from afar if the return is a random walk (Section 11.8), long before Hawking radiation would finish; or at a beat transition, if one comes sooner (Section 5.2). Within about a minute the traveler is gone as a pattern; in principle the traveler is never gone as a state.

9 Conservation and Consistency Conditions

Because $\nabla^\mu G_{\mu\nu} = 0$, the complete right-hand side of (1) together with the pressure sector must satisfy a compatible conservation law. With variable $\kappa(\phi)$, the simple condition $\nabla^\mu T_{\mu\nu} = 0$ is generally not sufficient by itself. The scalar equation (11) and the matter coupling must be chosen so that the full theory is diffeomorphism-consistent.

The revised Section 7 assists here rather than obstructing: because $T^{\text{matter}}_{\mu\nu}$ and $T^{\phi}_{\mu\nu}$ are now the two branches of a single field's excitations rather than two sectors joined by a step function, no discontinuous exchange term is introduced at the threshold.

9.1 Global regularity and the defocusing condition

A separate question, treated in earlier versions as a requirement that the medium remain smooth for all time, is whether it does. The framework cannot appeal to dissipation alone to secure this. Viscosity is conventionally expected to drain an energy cascade before it concentrates without bound, but that expectation is not a theorem: finite-time blowup constructions for the forced three-dimensional Navier–Stokes equations were announced in September 2026 and, while they remain subject to independent verification and do not transfer directly to a relativistic scalar medium, they remove the intuition that a viscous continuum is automatically globally regular. The shear and bulk viscosities $\eta(\phi)$ and $\zeta(\phi)$ of Section 2.3 therefore cannot be assumed to guarantee what they were implicitly relied upon to guarantee.

The mechanism available to this framework is not dissipation but the ultraviolet cutoff supplied by Section 7. The Ocean's amount cannot be bent over distances shorter than $\xi(\phi)$ without paying the cost of a dent: the gradient energy of the amount mode, together with its gap (Section 7), dominates the cascade at $k \sim 1/\xi(\phi)$ and arrests concentration at the healing length. This is the mechanism by which a laboratory superfluid remains regular where the corresponding classical fluid description does not.

That arrest is conditional on the sign of the self-interaction. Writing the relevant quartic term of the effective potential as

$$V_{\text{eff}}(\phi) \supset \frac{\lambda}{4}\left(\phi^2 - v^2\right)^2, \tag{24}$$

the medium must be *defocusing*, $\lambda > 0$: compression must raise the energy and be resisted. The defocusing case is energy-coercive and the cutoff at ξ holds. In the focusing case $\lambda < 0$ the self-interaction rewards further compression, the cutoff does not arrest the cascade, and concentration accelerates instead. The sign is therefore a stability requirement of the same standing as $L_{\chi} > 0$ and $\mathscr{K} > 0$, not a modeling preference. With the complex field as baseline the potential belongs to the modulus ρ (Section 2.2), and the argument is unchanged.

Regularity is a property of the state, not a guarantee. The sign of λ does not settle the matter for all times and places, because the stiffness that resists compression is the response of the medium *around the*

state it occupies. With the Lagrangian held fixed, that response differs between states. Even the quartic form above, with $\lambda > 0$, has curvature

$$V''_{\text{eff}}(\phi) = \lambda \left(3\phi^2 - v^2\right), \tag{25}$$

which is positive about the present vacuum $\phi = v$ and negative for $|\phi| < v/\sqrt{3}$. A medium driven into that region — by a beat transition, a contraction, or the core of a collapsing object — is no longer protected by the mechanism above. For this form the instability drives the medium back toward the stable state rather than into collapse; whether the full potential admits genuinely focusing states is open. The law is fixed; what varies is where the medium sits within it. No free function is added.

Four constraints follow. (i) Regions in different states are separated by boundaries that carry energy; walls of this kind within the observable universe are tightly excluded, so differing regions must lie beyond the horizon, in the past, or be separated without such walls. (ii) The microwave background is uniform to about one part in 10^5, which bounds how far the state could have varied across the observable universe at that epoch. (iii) A region that loses its stiffness collapses quickly and leaves relics, compact objects or primordial black holes, whose abundance is bounded. (iv) Where smoothness fails, the framework must state what follows — a bounce, a phase transition, or a genuine singularity at which it ceases to predict — and must tie each failure to a specified state of the medium, so that the possibility cannot be invoked case by case.

This section's argument rests on the framework's own structure. It does not depend on the blowup constructions cited above, which await independent verification.

A successful formulation should explicitly establish:

(1) covariance;

(2) total energy-momentum conservation;

(3) a well-posed scalar-field equation;

(4) absence of pathological ghosts or gradient instabilities;

(5) a stable low-density / weak-field limit;

(6) compatibility with precision tests of gravity and light propagation;

(7) a defocusing self-interaction about the present vacuum state, so that the healing-length cutoff of Section 7 arrests rather than accelerates concentration there, together with an account of the states in which it does not (Section 9.1);

(8) reversible underlying dynamics, with $\Pi_{\mu\nu}$ an effective, coarse-grained description (Sections 2.3 and 9.2).

9.2 Nothing is destroyed

The framework adopts a principle of preservation: the state of the Ocean is never destroyed. Formally this is reversibility, the requirement that each state determine its predecessor uniquely, which in quantum mechanics is unitarity. It does not require anything to run backward, and since time in this framework is the pace at which the Ocean re-makes its patterns (Section 5.1), "reversing time" is not an available operation. It is distinct from energy conservation: a process could conserve energy and still lose the state, which was the original concern about thermal Hawking radiation. The framework commits to both. Patterns are scrambled, coarse-grained disorder rises, and the arrow of time is that rise from a low-disorder start of the present beat; the fine-grained state is untouched. Recovery of a scrambled pattern need not mean running anything backward. In the spin echo, nuclear spins that have drifted out of step in an uneven magnetic field come back into step after a single pulse, each still turning forward at its own rate; only what has been truly randomized, by diffusion and by interactions between the spins, fails to return [28].

9.3 Where a wave's energy goes

The Ocean has no shore in space, so a wave crossing it is not stopped by a boundary. Its energy has five possible fates.

(1) **Spreading.** The wave's front grows and its energy is shared over it, with no absorption at all. This is the main fate of gravitational waves, to which the observable universe is nearly transparent.

(2) **Redshift.** Travel across an expanding region stretches the wave. The microwave background has lost over 99.9 per cent of its energy since it was released. General Relativity has no global energy conservation in an expanding spacetime and needs no account of the loss; this framework commits to preservation and must give one. Taking the divergence of equation (1) with $\nabla^\mu G_{\mu\nu} = 0$ fixes where the answer must appear: any transfer between radiation and the $P(\phi)$ sector is a non-zero divergence on the right-hand side, which would change how radiation density falls with expansion, and the microwave background and light-element abundances constrain that tightly. Until there is a calculation, redshift is the one fate of a wave the framework cannot follow.

(3) **Resonant absorption.** A pattern takes in energy only at its own frequencies, as an atom absorbs light only at its spectral lines, and passes the rest.

(4) **Cascade.** A large wave breaks into smaller ones and those into smaller still, until the energy reaches the healing length and is shed into the medium's background of short-wavelength excitations, as superfluid turbulence ends at the healing length. This is the pattern-level meaning of the viscous stress $\Pi_{\mu\nu}$.

(5) **Horizons.** A black hole absorbs any wave it intercepts, strips it, and returns its energy much later as Hawking radiation (Section 8.1).

The Ocean therefore has no shore in space but two in scale: the healing length below, where cascades end, and horizons above, which take in whatever reaches them.

10 Relation to Existing Work

The proposal that the vacuum is a superfluid, and that dark matter and dark energy are two behaviors of it rather than two substances, is not new. This section states what is already established, so that the framework's actual claim can be isolated from the parts of it that are inherited.

10.1 Two families of superfluid-vacuum programs

Existing work divides on whether spacetime geometry is fundamental.

Emergent gravity. Volovik's program [16] treats the quantum liquid as prior to spacetime: the metric, Lorentz invariance and gauge structure appear as low-energy collective behavior of the condensate. Zloshchastiev's logarithmic superfluid vacuum theory [29] derives an induced gravitational potential from the vacuum wavefunction and concludes that dark matter and dark energy are manifestations of the same object and therefore cannot be independent. That conclusion is the one reached here. The mechanism is not: Zloshchastiev distinguishes his approach explicitly from models that work by modifying the stress-energy tensor of the Einstein equations.

Volovik's program also supplies an argument this framework can use. Inhabitants of a superfluid, built from its own low-energy excitations, would perceive an effectively relativistic world and could not detect the underlying medium directly [16]. The same argument would account for the precision of Lorentz invariance at accessible energies here. It explains what cannot be observed; it does not exempt the framework from effects that are observed (Section 10, deficiency (4)).

Modified source. Huang and collaborators [1] treat a complex vacuum scalar field as a cosmic superfluid within standard General Relativity, identifying its energy density with dark energy and departures from vacuum density with dark matter. That work also treats galaxies as external sources perturbing the superfluid, and obtains the galactic halo together with the creation of vortices by galactic rotation — the closest existing precedent for the galactic sector of Section 6. Berezhiani and Khoury [2] construct an effective field theory in which dark matter is a superfluid with a galaxy-scale coherence length, described by phonons rather than, in the standard model's terms, collisionless particles, recovering MOND phenomenology on galactic scales. That model has since been disfavored by Milky Way data [12]: it over-predicts vertical accelerations of disk stars even while reproducing the rotation curve. Viscous cosmology [30] models the dark sector as a single fluid whose bulk viscosity produces a negative effective pressure and thereby mimics dark energy. The logotropic models of Chavanis [31] belong to the same family, yielding a density profile that falls as r^{-1} and offering an alternative to MOND that accounts for the Tully–Fisher relation.

On the cosmological side, the closest structural precedent is the cyclic model of Steinhardt and Turok [7] (Section 5). It, too, is a four-dimensional scalar field theory in which one field accounts for dark energy, the reversal from expansion to contraction, and the transition between cycles, and it couples that field to matter through a field-dependent factor. Its authors note that this coupling is a modification of General Relativity and produces a fifth force violating the equivalence principle unless the coupling's variation is very small at the present field value. The same constraint applies to $\kappa(\phi)$ here, and it is the first of the requirements of Section 2.5: every excitation must respond to the medium in exactly the same proportion.

Earlier versions placed this framework in the second family. With the position adopted in version 3.7 (the preface and Section 1), its account of what is fundamental moves to the first: the medium is primary and geometry is its effective description. Its working equation remains that of the second family, Equation (1), with the medium's stresses on the right-hand side.

10.2 What is inherited and what is not

Inherited: the identification of the vacuum as a scalar condensate; the proposal that the dark sector is one medium rather than two substances; the use of bulk and shear viscosity to generate an effective pressure; the appearance of a coherence length in the galactic regime.

Inherited also, and more directly than the preceding list suggests: the generation of galactic structure by vorticity in a vacuum superfluid. Huang and collaborators [1] obtain vortices from galactic rotation within a nonlinear Klein–Gordon description in which galaxies act as external sources. Section 6 reaches for the same phenomenon through different machinery — a dissipative stress tensor $\Pi_{\mu\nu}$ entering the field equation (1) rather than a source term in a wave equation. Whether these are distinct physical mechanisms or one mechanism in two formalisms is not settled here, and a credible version of this framework must say which.

The premise that ordinary matter is itself an excitation of the vacuum medium is also not new, and it is worth being precise about who holds it. Huang [1] and Berezhiani and Khoury [2] are two-species: the condensate constitutes the dark sector while baryonic matter is a separate species coupled to it. Superfluid vacuum theory is not. Its stated paradigm is a background quantum liquid of which "elementary particles", in its own terms, are excitations — one substance, as here.

Closest of all, on both what is fundamental and how gravity is recovered, is a line of Lorentz–Poincaré ether models outside the superfluid-vacuum literature. Arminjon's scalar ether theory [32, 33, 34, 35] takes a fluid ether as the only substance, with the elementary constituents of matter as organized flows in it; requires the ether's sound speed to be matter's speed limit, for that reason; makes clocks slow and rods contract in one ratio set by the local ether density; obtains gravitational waves from the ether's compressibility; and replaces the singularity of collapse with a bounce. Its isotropic version has the exponential metric, agrees with General Relativity at first post-Newtonian order, and does not produce frame dragging. Before it, Podlaha, Sjödin and Broekaert treated the ether's density as a refractive

index and scaled rods, clocks and light together; Broekaert reproduces the four classical tests at first post-Newtonian order with the scaling this framework requires (Section 11.5) [36].

What remains distinctive is narrower than earlier versions claimed. Superfluid vacuum theory obtains a single substance by making geometry emergent from the condensate, and this framework now takes the same position on what is fundamental, so that is no longer a point of difference. Nor is the refractive route: the ether models above recover gravity's classical tests through a medium whose state scales rods, clocks and light together. Five elements remain. Four concern gravity. The carrying speed varies with the state, and that variation is gravity (Section 11.1), where Arminjon holds the ether's sound speed fixed; and no fixed speed remains anywhere in the law. Near a mass it is the Ocean's rulers and clocks that are squeezed, not the Ocean, and patterns fall by refraction rather than by a pressure push. The candidate mechanism of Section 11.6, a current along the time direction slowed by drag and kept in step, has not been found in the literature reviewed. And the principle that the Ocean's field is not its own source selects General Relativity's geometry, with a point of no return, over the exponential metric that the ether models share (Section 11.7). The fifth is the mechanism of the partition, the division of one field's excitations into two branches (Section 7), the amount mode (localized, $T_{\mu\nu}^{\text{matter}}$) and the phase ripples (coherent, $T_{\mu\nu}^{\phi}$), with the threshold the amount mode's gap, Equation (19). The consequence developed in Section 9 — that no discontinuous exchange term between the two is required, because both are populations of one field — follows from that construction. It removes an obstruction to the conservation conditions; it does not by itself establish them, which remains outstanding.

10.3 The electron-scale coincidence

The numerical agreement of Section 7 has been approached from the opposite direction. Working from a logotropic equation of state, Chavanis [31] recalls an expression for the cosmological constant in terms of the electron mass and the fundamental constants,

$$\Lambda = \frac{G^2 m_e^6}{\alpha^6 \hbar^4} \approx 1.36 \times 10^{-52}\,\mathrm{m}^{-2}, \tag{26}$$

a relation of Eddington type. There the electron mass fixes Λ; here the electron mass is the readout of $\xi(\phi_0)$ for the present beat. Both routes tie the electron scale to a vacuum-scale quantity, from opposite ends.

Two cautions attach to this. The relation is *recalled* rather than derived in that work, so the primary source lies earlier in the logotropic literature. And independent arrival at a coincidence is weak evidence that it is not accidental; it is not this framework's discovery, and the open question of Section 7.7, item (6), remains untouched by it.

10.4 Known deficiencies relative to current practice

Points where this document is behind the literature it should be engaging, or exposed to measurements it has not yet addressed.

(1) **First-order dissipative hydrodynamics.** The constitutive form of Section 2.3 is of Eckart–Landau type, which is known to be acausal and generically unstable in relativistic settings [37]. Current viscous cosmology uses causal second-order formulations of Israel–Stewart type [38]. A viable version of this framework must do the same.

(2) **Vertical dynamics.** Section 6's decomposition is radial. Milky Way kinematics constrain the two components separately, and they do not want the same thing: the baryonic distribution supplies too little gravity to account for the rotation curve but enough to account for the vertical velocity dispersions near the Sun, so substantial enhancement is required radially and essentially none vertically [11, 12]. Superfluid dark matter fails this test because its phonon-mediated force is approximately parallel to the baryonic gravitational field and therefore enhances both components by the same factor [12]. Cold dark matter passes because its vertical acceleration is dominated by

baryons, with the halo subdominant. The resulting criterion applies to any model: the enhancement of vertical acceleration must be suppressed relative to the radial enhancement. Whether this framework is exposed depends on whether $v_{\phi,\mathrm{eff}}$ tracks the baryonic acceleration field — in which case it inherits the parallelism problem — or constitutes an independent source. Section 6 proposes the second, shear bound to the galaxy itself. Under the split of Section 11, that shear is a source of the geometry, not a force on stars. Slow patterns respond mainly to the slowing of their rhythm (Section 11.5), which follows the overall amount of a source more than its direction. The lattice's anisotropy therefore does not by itself give an anisotropic pull: what must be shown is that its effective source is distributed so that the resulting acceleration is enhanced radially and not vertically. Section 6 now gives one form that does, a swirl of the Ocean around the galaxy that is the same at every height near the disk; the full confrontation with the data is not done.

(3) **The status of $c(\phi)$.** A variable effective propagation speed is natural in the emergent-gravity family, where the metric is acoustic. The refractive reading is now the framework's account of gravity (Section 2.5), which places it with that family on what is fundamental and incurs the family's chief obligation: to derive the effective geometry rather than assume it. Section 11.1 settles the reading: $c(\phi)$ is the single local carrying speed of every disturbance, the refractive reading, and not a second causal speed; its two readings, the same on the spot and varying from afar, are separated there. What remains is to derive it.

(4) **The vacuum is already measured at the electron scale.** In quantum electrodynamics, vacuum polarization screens every charge out to the electron's reduced Compton wavelength — the same 386 femtometers as ξ_0 — and contributes measurably to the Lamb shift and to the electron's anomalous magnetic moment, in agreement with theory to many significant figures. The vacuum also behaves as a nonlinear optical medium. Light-by-light scattering, forbidden in classical electrodynamics, has been observed in ultra-peripheral heavy-ion collisions [39], and the strong linear polarization of optical light from the neutron star RX J1856.5−3754 is evidence for vacuum birefringence in a strong magnetic field [40]. The Casimir effect is measured. If ξ_0 is a property of the Ocean, these are the Ocean's responses at and near its healing length, and the framework must reproduce them quantitatively. Until it does, the electron-scale coincidence of Section 7 is matched by an obligation of the same order.

(5) **Antimatter is measured.** Every charged localized excitation has an antimatter partner, observed directly: the positron [41], the antiproton and antineutron, antinuclei up to antihelium-4, and bound antihydrogen, trapped and laser-cooled at CERN [42]. Matter and antimatter agree to extreme precision: the antihydrogen 1S–2S transition matches hydrogen to about 2 parts in 10^{12} [43]; the antiproton charge-to-mass ratio matches the proton's to about 16 parts in 10^{12} [44]; antihydrogen falls toward the Earth, measured at $0.75\,g$ with an uncertainty of about $0.2\,g$, consistent with g, and repulsive gravity is ruled out [45]. The one measured difference is CP violation in the weak interaction, seen in kaons, B and charm mesons and, in 2025, baryons [46], and too small in the Standard Model to explain the excess of matter.

The framework must therefore produce, for each localized excitation, a partner of identical mass and magnetic moment and opposite charge, equal response to the slowing of the Ocean (the first of the three decisive requirements of Section 2.5), and room for the small CP asymmetry. A single real scalar field cannot: its excitations are their own antimatter partners. This is a further reason for the complex-field formulation of Section 2.2, and the most fundamental of them.

Antimatter also tests the preferred frame. In standard field theory a violation of CPT symmetry implies a violation of Lorentz invariance, and a medium with a rest frame is a natural source of both. The comparisons above are therefore among the most precise existing limits on the Ocean's preferred frame. Antimatter is not a separate obligation from Section 7.7, item (3); it is one of the instruments that tests it. One check arises from the complex field itself: in a state with nonzero phase rate μ, excitations of opposite charge under the Ocean's $U(1)$ differ in energy by

2μ (Section 11.4); if antimatter partners are such charge-conjugate excitations, these comparisons bound it.

10.5 The structural question

If the vacuum is a medium possessing density, pressure and a rest frame, it is reasonable to ask why it requires a background spacetime to occupy. The emergent-gravity family answers that it does not, and that geometry is the medium's own collective behavior. Earlier versions of this framework posited a medium and a geometry as distinct structures coupled through $\kappa(\phi)$. Since version 3.7 the framework answers the question in favor of the medium. There is no background spacetime for the Ocean to occupy; the geometry of Equation (1) is how the medium's state is experienced by excitations of the medium, and $\kappa(\phi)$ expresses how strongly that state responds to concentrated energy. The position is now stated. It has not yet been derived, and deriving the effective geometry from the dynamics of the medium is the task it creates (Section 12). The derivation needs a strict split, or the medium is counted twice. Section 11 states the split, the principle it rests on, and why the present form of the law cannot yet carry it.

11 One Speed and the Split

Section 10 ends with the task the medium-first position creates: deriving the effective geometry of Equation (1) from the medium, with a strict split between the background that defines the geometry and the excitations that act as sources. This section states the principle on which that split rests, shows why the minimal form of the Ocean's law cannot carry it and where the fixed speed hiding in it lies, states the condition the weak-field limit imposes, offers a candidate mechanism, and derives from two of the framework's principles the geometry around a non-spinning mass, with its point of no return. It does not derive Equation (1) in general; that remains the central task of Section 12.

11.1 One speed

The Ocean has one speed: the speed at which it carries a disturbance, written c_s and called here the carrying speed. Everything runs at it — light, the internal workings of every pattern, every signal. The Ocean's own flow is not a disturbance traveling through the Ocean, and has no such limit; "faster than c_s" is always measured against some other part of the Ocean, never against the water a pattern sits in. Where the flow outruns c_s there is a horizon: disturbances still move at c_s through the local medium, but the medium carries them away faster than they can make headway.

What is called the speed of light is therefore not a constant of nature but the Ocean's local carrying speed. It is always measured with the same value because rulers and clocks are themselves patterns running at that speed; a change in c_s is invisible locally by construction. Gravity is where c_s varies from place to place.

There is therefore no fixed c. Every place a speed appears in the Ocean's law, it is c_s, the Ocean's own carrying speed where it is. The law is written on a single field of rulers and clocks that is part of the Ocean's state, not a stage, and every term uses it: the Ocean's own ripples, every pattern, light. The carrying speed then has two readings. On the spot, every pattern measures the same value, because its rulers and clocks are made of the same Ocean. Seen from afar, it varies, and that variation is gravity. For the Ocean's own ripples to travel at the one speed, the Ocean must be stiff: in the minimal law of Section 11.4 the shortfall of the ripple's speed is $a = 2\mu^2/(\lambda v^2)$, and the coefficient of the k^4 term is about $4a^2$, so stiffness removes both the mismatch of speed limits and the short-ripple overshoot.

There is one speed because there is one substance. A disturbance passes from place to place only by the Ocean handing it along, and the speed of the hand-off is set by two properties of the medium, how stiffly it resists being bent or compressed and how much inertia it has: c_s^2 is their ratio. At any place there is one stiffness and one inertia, and so one speed. A second speed would require a second substance, or a layer running beneath the Ocean, which would be the inherited stage returning.

A localized pattern is a ripple circulating at c_s, confined. At rest all of its carrying speed goes into circulating. In travel part of it carries the pattern forward, and less remains for its internal rhythm, which slows by $\sqrt{1-v^2/c_s^2}$. Relativistic time dilation is then a consequence of the medium, and a pattern can never reach c_s, because nothing would remain to circulate. This is the dynamical reading of relativity first proposed by Lorentz and FitzGerald. The step from "matter is flows in the medium" to "the medium's sound speed is matter's speed limit" has a precedent: Arminjon's ether theory requires the ether's sound speed to equal c for exactly this reason [34]. There that speed is fixed everywhere; in the Ocean it varies with the state, and its variation is gravity.

11.2 The rules are fixed

The Ocean's law is the same everywhere. What differs from place to place is the state a pattern sits in. Every pattern responds to the Ocean around it in two ways: it is carried by currents, and it is refracted by differences of state. Neither is felt from inside: a pattern riding even the swiftest current notices nothing, and its internal workings run as they would anywhere. This is the equivalence principle in the Ocean's terms.

The response may depend on where a pattern is and what the Ocean is doing there, never on what the pattern is made of. Tightly and loosely wound patterns at the same place are carried and refracted identically; composition dependence of free fall is excluded at the level $\eta(\mathrm{Ti},\mathrm{Pt}) = [-1.5 \pm 2.3\,(\mathrm{stat}) \pm 1.5\,(\mathrm{syst})] \times 10^{-15}$ [47].

What breaks a pattern is not the speed of a current but its change across the pattern. A uniform current carries everything along unfelt; a pattern is torn when its front lies in faster flow than its back. A loosely bound pattern spans more of the Ocean and holds itself together more weakly, so it is torn sooner and farther out: stars before people, people before atoms, atoms before nuclei. That is tidal disruption (Section 8.2).

11.3 What sits on each side of the field equation

The background state of the Ocean sets two things: the local carrying speed $c_s(x)$ and the flow $u^\mu(x)$. The effective geometry from which $G_{\mu\nu}$ is built is constructed from these alone, in the acoustic or river form. The background's own pressure $P(\phi)$ belongs to this side, as part of what sets c_s; it is not a source. The right-hand side of Equation (1) holds only what sits above the background: localized patterns ($T^{\mathrm{matter}}_{\mu\nu}$), coherent ripples ($\delta T^{\phi}_{\mu\nu}$), and the coarse-grained vortex stress $\Pi_{\mu\nu}$. Nothing is counted twice. The geometry is the Ocean's own field of rulers and clocks (Section 11.1).

A geometry built from c_s and the flow alone fixes the light cones, and with them the paths of light. It cannot by itself fix the overall conformal factor that governs slow patterns: an acoustic geometry can match the Schwarzschild geometry only up to such a factor [48]. In the Ocean that factor is supplied by how patterns scale with the state (Section 11.5).

11.4 Why the minimal law cannot carry the split

Two results, computed for this version and checked numerically, show that the simplest law consistent with Section 2.2 cannot deliver the split as it stands. That law, with the quartic potential of Section 9.1, is

$$L = -(\partial\rho)^2 + \rho^2\mu^2 - \frac{\lambda}{4}\left(\rho^2 - v^2\right)^2, \qquad \mu^2 \equiv -\nabla_\alpha\theta\,\nabla^\alpha\theta, \tag{27}$$

with μ the phase rate of Equation (6). At long wavelength the modulus sits at $\bar{\rho}^2 = v^2 + 2\mu^2/\lambda$, leaving a single-field superfluid with pressure $P(\mu^2) = v^2\mu^2 + \mu^4/\lambda$, whose derivative $P' \equiv dP/d(\mu^2) = \bar{\rho}^2$ is exactly the modulus squared.

The phonon geometry gives the wrong light bending. Splitting the phase into background and ripple, $\theta = \bar{\theta} + \delta\theta$, the ripple propagates in the disformal acoustic metric [49, 48]

$$g^{\text{eff}}_{\mu\nu} = \Omega^2 \left[g_{\mu\nu} + \left(1 - c_s^2\right) u_\mu u_\nu \right], \qquad c_s^2 = \frac{P'}{P' + 2\mu^2 P''}, \qquad \Omega^2 = \frac{P'}{c_s}. \tag{28}$$

With $a \equiv 2\mu^2/(\lambda v^2)$, $c_s^2 = (1+a)/(1+3a)$. In the medium's rest frame $|g^{\text{eff}}_{00}| = P' c_s$ and $g^{\text{eff}}_{ij} = (P'/c_s)\,\delta_{ij}$, so the post-Newtonian parameter that measures the curvature of space per unit slowing of time is

$$\gamma = -\frac{2+3a}{3a}, \tag{29}$$

always below -1: space contracts where clocks slow, the opposite of General Relativity's $\gamma = 1$. For any single-field superfluid of this kind, $\gamma = 1$ would require P' to stay fixed while c_s varies, which is impossible, since both move with μ^2 alone. In the stiff limit $a \to 0$ the time part does not change at first order, so light bends but slow patterns feel no pull; at large a the metric is conformal and $\gamma \to -1$, with no light bending. With the modulus held rigidly at v, $P'' = 0$ and the ripple sees flat space. Matter and light cannot both be reading the acoustic geometry of the phase.

Light and matter have different speed limits. In units where the speed of the flat background in Equation (27) is one, and in the frame turning with the phase, the small disturbances satisfy

$$\left(\omega^2 - k^2\right)\left(\omega^2 - k^2 - M^2\right) = 4\mu^2\omega^2, \qquad M^2 = \lambda\bar{\rho}^2. \tag{30}$$

A gapless branch travels at the c_s above. A gapped branch, the candidate for matter, has rest rhythm $\omega_0^2 = \lambda v^2 + 6\mu^2$. At high k both branches approach $\omega = k \pm \mu$: their speed limit is set by the flat background written into the law, not by the Ocean's state. Two consequences follow. A pattern's rhythm times its size equals its own speed limit, which here is fixed, so where its rhythm slows its size must grow; clocks and rulers move in opposite directions, and the lockstep of Section 11.5 cannot occur. And wherever the Ocean is not fully stiff, light is slower than matter's limit, so fast charged patterns would outrun light and radiate it away. Cosmic-ray protons near 10^{20} eV bound such a difference to about one part in 10^{23} [50], and for electrons to about one part in 10^{15} [51]. Since $1 - c_s \approx a$ for small a, the Ocean under this law would have to be stiff to $a \lesssim 10^{-23}$, where c_s stops responding to the state and gravity disappears.

The diagnosis: Equation (27) is written on a flat background, the stage removed in version 3.7. Every term of the form $(\partial\Phi)^2$ means $(1/c^2)(\partial_t\Phi)^2 - (\nabla\Phi)^2$ with c a constant of that stage, and the fixed speed appears in three places: in how matter's inner workings move (the mismatch above), in the short-ripple overshoot of the k^4 term, and in the law's own bookkeeping, where $c_s^2 = dp/d\varepsilon$ is measured against it. The last is why the weak-field condition of Section 11.5 and the equation of state used in version 3.9 for Section 11.6 contradicted each other at first order: one is the on-the-spot reading of c_s, the other the reading from afar. With the law written on the Ocean's own rulers and clocks (Section 11.1), stiffness, $a \lesssim 10^{-23}$, is required rather than fatal: gravity is the change of the rulers and clocks seen from afar, not of the on-the-spot ripple speed. Writing that law out in full is the central research problem (Section 12).

11.5 The weak-field condition

Near a mass c_s is slower. Light bends because c_s varies along its path. Slow patterns fall because their internal rhythm is slower deeper in, and a pattern drifts toward where its rhythm, and so its energy, is lower: refraction for matter. The one-speed principle ties a pattern's rhythm and size together, rhythm times size equaling c_s, which makes the change invisible locally. It leaves open how the slowing is shared. The standard post-Newtonian account settles it: the equivalence principle alone gives half the observed deflection of starlight, and the curvature of space, measured by γ, gives the other half [52]. General Relativity's light bending therefore requires an even split,

$$\text{rhythm} \propto \sqrt{c_s}, \qquad \text{size} \propto \sqrt{c_s}. \tag{31}$$

The same scaling — a coordinate light speed falling as Φ^2, rods and clocks each scaling as Φ, inertia as Φ^{-3} — was used by Broekaert, after Sjödin, to reproduce the four classical tests at first post-Newtonian order [36]; Equation (31) is that condition stated for the Ocean. Large bodies follow automatically: a star is made of patterns, and if each scales alike the star falls as its parts do. The condition is a rule for the weak field: close to a mass it gives way, and Section 11.7 gives what replaces it. A route through the Ocean's inertia, by which a rising phase rate near a mass would produce the scaling through a laboratory-superfluid healing length, fails: that formula does not carry over to the Ocean, and combined with the version 3.9 equation of state it contradicts the Ocean's thermodynamics at first order (Section 11.4). Equation (31) enters only as one constant, $\gamma = 1$, and is otherwise not needed.

11.6 A candidate mechanism: the current through time

This subsection is a candidate, not a derivation. Ordinary sideways flow past a star cannot be the source of its gravity: the Sun travels at about $370\,\mathrm{km\,s^{-1}}$ relative to the microwave background, and gravity from flow past it would be lopsided, stronger on its flanks than ahead and behind, which the post-Newtonian preferred-frame parameters exclude at $\alpha_1 \lesssim 10^{-4}$ and $\alpha_2 \lesssim 2\times10^{-9}$ [52]. One current is the same whichever way a star travels: the flow that carries every pattern forward in time. On the one-speed principle every pattern moves through the Ocean at c_s in total, all of it along time when at rest.

(i) **The star slows the current.** A star exists at every moment, so it lies along the current through time rather than across it, and a frictionless flow along such a body is undisturbed. The star must therefore drag on the current, through the medium's viscosity. The slowed current spreads sideways through three space directions, so the deficit falls off as $w \propto M/r$: Newton's inverse distance, with a strength set by the drag per pattern relative to the Ocean's viscosity.

(ii) **Clocks slow by w.** Patterns are carried through time by the local current, so their rhythm slows by w. This is the half of gravity that makes things fall.

(iii) **The Ocean stays in step.** A river speeds up past an island because the island blocks the water and the same amount must squeeze past its flanks. A star blocks nothing: it is made of the Ocean, and it lies along the current, which has always flowed past it. What the Ocean must not do is fall out of step with itself. If the Ocean near a star turned more slowly, seen from afar, than the Ocean far away, the mismatch would grow at every moment, and a growing mismatch is a growing current: an upstream wake, which nothing observed supports. So, seen from afar, the Ocean's phase turns at one rate everywhere; it is one clock for the whole Ocean. Measured on the spot, where clocks run slow, it turns faster, by exactly the clocks' slowing, $\mu_{\mathrm{local}} = \mu_\infty/U$ with U the rate of clocks seen from afar. This is the relativistic form of Bernoulli's relation for a fluid at rest in a gravitational field: Klein showed that its chemical potential, including the mass energy, times $\sqrt{|g_{00}|}$ is the same everywhere [53], and the Ocean's chemical potential is its phase rate (Section 2.2). On the spot the current through time is always at the carrying speed, so all change shows in the view from afar.

(iv) **The amount of Ocean stays the same.** The Ocean is stiff (Section 11.1): its amount ρ barely gives. Near a mass it is not the Ocean that is squeezed but its rulers and clocks. The pressure–density law that version 3.9 needed for this step, of generalized Chaplygin form [54, 55], is no longer required.

(v) **Far from a mass: clocks and rulers in lockstep.** The three rules above fix clocks and the Ocean's turning, but not rulers. Far from a mass the lockstep of Equation (31) supplies them, and to first order

	General Relativity	Candidate
Clock rhythm	$1-w$	$1-w$
Light speed seen from afar	$1-2w$	$1-2w$
Pattern size ($= c_s/\mathrm{rhythm}$)	$1-w$	$1-w$

Write the rate of clocks seen from afar as U, so that $|g_{00}| = U^2$, and $u \equiv GM/(rc_0^2)$. If each layer of the Ocean subtracted a fixed amount from the current, $U = 1-u$, the post-Newtonian parameter β would be $\frac{1}{2}$ and Mercury's perihelion would advance $(2+2\gamma-\beta)/3 = 7/6$ times too fast. If each layer slows the current by a fraction of what it receives, $U = e^{-u}$ to second order and $\beta = 1$. Rulers set instead by "the amount looks fixed from afar" would give $\gamma = -\frac{1}{3}$, and fail light bending.

(vi) **Close to a mass, lockstep gives way.** Taken all the way in, lockstep gives the exponential metric of Yilmaz, $|g_{00}| = e^{-2u}$, $g_{ij} = e^{2u}\delta_{ij}$, which Arminjon's isotropic ether theory has in full [33] and which underlies Broekaert's model [36]. In it outward travel slows without limit but never stops: there is no point of no return, and the shadow of a black hole would be 4.6 per cent larger than General Relativity's, outside one of the two ranges the Event Horizon Telescope allows for Sagittarius A* [27]. A horizon is a point of no return: before it escape is possible, after it it is not. The Ocean needs one, so lockstep is a rule for the weak field only. Section 11.7 gives the rule that replaces it close in. The horizon at $2.9\,GM/c_0^2$ given in version 3.9 is withdrawn: it came from applying water's form of Bernoulli's relation along the time direction, which the rule of item (iii) replaces.

The candidate's inputs are now the first-order lockstep constant, equivalent to the observed bending of light, and the strength of the drag, which sets G. The composition of the slowing in layers, an input in version 3.9, follows from the rules of Section 11.7. One cosmological requirement stands on its own: unified fluids whose pressure is a function of one density are excluded as the cosmic medium once $|d\ln p/d\ln n|$ exceeds about 10^{-5}, because their sound speed erases the growth of structure on galactic scales [56]. The Ocean has two state variables, the amount ρ and the phase rate μ (Section 2.2), so its cosmological pressure need not be such a function; that it is not is to be shown.

11.7 The strong field and the point of no return

Two principles the framework already holds fix the geometry around a non-spinning mass completely. Write it in isotropic form,

$$ds^2 = -U^2c_0^2dt^2 + \psi^4\left(dr^2 + r^2d\Omega^2\right), \tag{32}$$

with U the rate of clocks seen from afar and ψ^2 the number of the Ocean's own rulers that fit in a unit step of r.

(a) **Staying in step, counted with the Ocean's own rulers.** The slowing of clocks spreads from the mass by the inverse-distance law, with distance counted by the Ocean's own rulers, which shrink near the mass, so that more of them fit across a given span and $\psi^2 > 1$, and not by a flat ruler underneath (Section 11.1): $\nabla\cdot(\psi^2\nabla U) = 0$.

(b) **Nothing counted twice.** Where there is no pattern, the Ocean around a mass does not squeeze its own rulers; the field is not its own source (Section 11.3): $\nabla^2\psi = 0$.

With the first-order lockstep used only to fix one constant, $\gamma = 1$, the solution is

$$\psi = 1 + \frac{GM}{2rc_0^2}, \qquad U = \frac{1-GM/(2rc_0^2)}{1+GM/(2rc_0^2)}, \tag{33}$$

General Relativity's geometry outside a non-spinning mass. The consequences, each checked numerically for this version:

- **The point of no return.** U reaches zero, seen from afar, at isotropic radius $GM/2c_0^2$, areal radius $2GM/c_0^2$: General Relativity's horizon, now produced by the geometry rather than assumed. It is the same point for every pattern, since every pattern has the same carrying speed.
- **Mercury, derived.** To second order U agrees with e^{-u}, so $\beta = 1$ follows; the slowing composes in layers as a consequence, not an input.

- **The shadow.** Light orbits at areal radius $3GM/c_0^2$ and the shadow's radius is $3\sqrt{3}\,GM/c_0^2$, as in General Relativity. The Event Horizon Telescope finds the shadow of Sagittarius A* within about ten per cent of that value: a fractional deviation of $-0.04^{+0.09}_{-0.10}$ or -0.08 ± 0.09 for the two determinations of its mass and distance, and no evidence of a surface [27]. The orbit of the star S0-2 about the same black hole gives $(2+2\gamma-\beta)/3 = 1.1 \pm 0.19$ (reported there); the Ocean gives one.
- **What separates the Ocean from the nearest precedents.** The exponential metric of Section 11.6 satisfies rule (a) but not rule (b): its empty space squeezes its own rulers, which counts the medium twice. Rule (b) is the split of Section 11.3, and it is what selects a point of no return.

The derivation assumes a static, spherically symmetric mass and the isotropic slicing, which spherical symmetry always allows. A spinning black hole needs the Ocean's account of frame dragging, which is not given; the Kerr shadow varies by about eight per cent with spin and viewing angle [27].

11.8 At the edge and inside

Nothing happens at the edge. If the Ocean stayed in step while holding still all the way down to the point of no return, it would there have to turn infinitely fast, and a traveler falling in would meet it turning ever faster. It does not need to. In the stiff Ocean the phase obeys the wave equation on the geometry of Equation (33), and that equation has a solution that is in step far away and smooth across the point of no return. With r now the areal radius, the one a sphere's surface measures (not the isotropic radius of Equation (33)), and in units with $G = c_0 = 1$,

$$\theta = -\mu_\infty [t + f(r)], \qquad r^2 \left(1 - \frac{2M}{r}\right) f'(r) = 4M^2. \tag{34}$$

Far away $f' \to 4M^2/r^2$ and the phase turns at one rate seen from afar; at the point of no return the phase gradient stays finite for a falling traveler. Near the edge the Ocean keeps in step by flowing gently inward, not by turning faster. The traveler falls with it and meets nothing there, as the equivalence principle requires. The Ocean is whole and connected across the point of no return; what is cut off is outward travel. The inflow carries energy in at a rate proportional to $\mu_\infty^2 M^2$: the black hole takes in the Ocean's background very slowly, at a rate set by its pressure's departure from $w = -1$.

Inside. Past the point of no return, forward in time points inward, and the Ocean's own phase flows inward with whatever falls. A pattern's time is its rhythm; as patterns are stripped, their clocks go with them.

The core: flow-through. Nothing is gathered at the center. Recycling, for the Ocean, is flow-through: the inward flow does not stop or collect at the center, but passes through and returns to the Ocean, later. Time is recycled with it: inward on the way in, outward again on the way out. A black hole and a star are the two halves: a star sends energy outward from a dense core, and the return from a black hole runs the same way. Two results bear on how.

- **Stiffness cannot turn the flow around.** For any material with $\varepsilon + p \geq 0$, which includes the stiffest that one speed allows, the rate of change of a collapsing region's expansion is $dH/dt = -4\pi G(\varepsilon + p) \leq 0$; once collapse is under way inside the point of no return it cannot reverse. The Ocean's resistance to compression ($\lambda > 0$, Section 9.1) is not enough.
- **The Ocean's grip can.** Equation (1) carries $\kappa(\phi)$, how strongly the Ocean's rulers and clocks respond to what is in them. If that grip loosens under extreme compression, $G_{\text{eff}} = G(1 - \rho/\rho_c)$, the collapsing interior obeys $H^2 = (8\pi G/3)\rho(1 - \rho/\rho_c)$: collapse slows, stops at ρ_c and reverses. In a ball of patterns falling in from rest far away, the time from crossing the point of no return to the turnaround, on a clock falling with it, is about 10^{-4} s for a black hole of ten solar masses,

28 s for Sagittarius A* and 12 hours for M87*; nothing is crushed to a point. Neutron stars exist and black holes of a few solar masses form, so $\rho_c \gtrsim 10^{18}\,\mathrm{kg\,m^{-3}}$. If ρ_c is set by a length ℓ through $\rho_c c_0^2 \sim \hbar c_0/\ell^4$, then $\ell \lesssim 0.8$ femtometers, some five hundred times finer than the healing length: the turnaround is set by structure below ξ_0 (Section 11.9).

The return, seen from afar. On a falling clock the turnaround is quick; for an outside observer the delay is set by the region just outside the point of no return, through which the reversed flow must come back. That region depends on how the Ocean's grip loosens there, which is a property of what the Ocean is made of. Written as a function of the fine scale ℓ, the delay is $\tau \sim (GM/c_0^2)^{n+1}/(c_0\,\ell^n)$, with $n = 1$ if the loosening builds up quickly near the edge and $n = 2$ if slowly; both forms are borrowed scalings, to be derived. The $n = 1$ form is a random walk across the black hole's size in steps of ℓ, the same rule that keeps light made in the Sun's core inside the Sun for tens of thousands of years or more; what comes out is not what went in, as gamma rays made in the core leave the surface as sunlight. Black holes of a few solar masses are seen merging after billions of years, so with $n = 1$ the step must be $\ell \lesssim 6 \times 10^{-19}$ m, close to the scale, about 10^{-18} m, down to which the electron shows no size; with $n = 2$ any ℓ below 0.8 femtometers passes. For Sagittarius A* the return then takes of order 10^{22} years or longer. Every known black hole returns what it holds far in the future, consistent with none having been seen to do so. The release, if a random walk, would leak out gradually and thermalized, like starlight, rather than as a single burst.

11.9 What the Ocean must be made of

This version does not say what the Ocean is made of. Any substrate must satisfy four requirements: a single speed, so that no signal passes through it faster than c_s, which would return the stage one level down; a grain, if there is one, hidden below every tested scale; internal structure that can be twisted; and the ability to hold confined, circulating patterns.

The third follows from light. Light's oscillation is entirely transverse, in two independent directions; crossed polarizers extinguish it. A complex field is one complex number at each point, a size and a phase, and a number has no direction, so every free ripple of it is sound-like and could not be polarized. If light is a ripple of the Ocean, each point must carry something with a direction that can be twisted, and under the one-speed principle its twisting and compressional ripples cannot travel at different speeds, as they do in a solid. The same gap appears twice more: gravitational waves are transverse, in two patterns, and spin-$\frac{1}{2}$ patterns need internal structure a single number cannot carry (Section 7.7, item (7)). Three independent demands point at one change, an Ocean with more inside each point than a size and a phase.

Black holes add two numbers (Section 11.8). The substrate must turn collapse around at a density of about $10^{18}\,\mathrm{kg\,m^{-3}}$ or more, which points to structure at about 0.8 femtometers or finer, well below the healing length. And if the Ocean's grip loosens quickly near a point of no return, the substrate must redirect flow at spacings of about 6×10^{-19} m or less. Its account will fix how long a black hole takes to return what it holds.

12 Research Program and Testable Predictions

- Derive the field equations from a covariant action rather than introducing stress terms phenomenologically.
- Derive the effective equation of state $P(\phi)$ and determine whether a stable oscillatory cosmological solution exists.
- Solve the weak-field, stationary, rotating-galaxy limit and compare predicted rotation curves with baryonic distributions.
- Determine whether the same parameter set fits multiple galaxies without an independently chosen halo profile.

- Calculate the weak-field and post-Newtonian limits to verify recovery of Solar-System observations.
- Show that scalar, gravitational, and electromagnetic perturbations all travel at the one carrying speed $c_s(\phi)$, as Section 11.1 requires: the refractive reading of $c(\phi)$ is adopted, and what remains is to derive it.
- Test cosmological expansion, structure growth, gravitational lensing, and background-radiation constraints.
- **Derive the gap of the amount mode**, and with it the healing length $\xi(\phi)$ of (17), from the scalar Lagrangian.
- **The beat's constant.** Determine what at a beat transition sets the electron's scale for the beat, and how the change of state that drives expansion keeps it steady (Section 7, Consequences). This replaces the earlier task of confronting a predicted drift in ξ.
- **Resolve or explain the hierarchy** between the mass gap Ω_{gap} required by galactic dynamics and the stiffness scale implied by $\xi_0 \approx 386$ femtometers. The candidate route of Section 7.7, item (5), divides this into three tasks.
  - *Protection.* Write the scalar sector explicitly in the complex field (5), with P, κ, ξ, η, ζ and the quartic potential of Section 9.1 as functions of the modulus or of X, and determine whether the shift symmetry of the phase survives the matter coupling and the dissipative sector. If it does not, compute the gap that results.
  - *Size.* Determine what fixes the extent of a galactic solution, given that the medium supplies no kiloparsec scale of its own: which features of the full state of the disturbance set it — energy, organized and random flow, exchange with the surrounding medium — and where their values originate.
  - *Rotation curves without per-galaxy freedom.* Derive $v_{\phi,\text{eff}}(r)$ from the baryonic distribution alone, with no per-galaxy freedom, and compare with the observed radial acceleration relation [10, 9], including its universal acceleration scale $g_\dagger \approx 1.2 \times 10^{-10}\,\text{m}\,\text{s}^{-2}$, which the framework must then account for, and extend the derivation to pressure-supported galaxies, which follow the same relation [9]. This must be done jointly with the vertical-acceleration item below, since a contribution fixed by the baryons is the kind most exposed to it.
- **Determine the sign of λ** from the chosen $L(\phi, X)$ and verify the defocusing condition of Section 9.1. A framework whose potential turns out to be focusing about the present vacuum has no mechanism arresting concentration and is not viable as written. Map the states in which stiffness is lost, determine what the medium does there, and confront the result with domain-wall, microwave-background and relic-abundance bounds.
- **Distinguish the galactic mechanism from its precedent.** Determine whether the rotational support of Section 6, generated by $\Pi_{\mu\nu}$, is physically distinct from the vortex formation obtained by Huang and collaborators [1] from a nonlinear Klein–Gordon equation with galaxies as external sources, or whether the two are the same mechanism expressed in different formalisms.
- **Establish the direction of the vacuum contribution.** Determine whether the acceleration produced by $\Pi_{\mu\nu}$ is parallel to the baryonic gravitational field, as a scalar-mediated MOND-like force is, or whether its anisotropic shear components yield a contribution with an independent orientation. This is the question on which the galactic sector turns: a parallel contribution enhances radial and vertical accelerations by a common factor and is excluded by local Milky Way data [11, 12], whereas an independent, quasi-spherical contribution is not. Then compute the predicted vertical acceleration explicitly and confront it, together with the rotation curve of Section 6, with stellar kinematics.

- **Reformulate the dissipative sector causally.** Replace the first-order constitutive relation of Section 2.3 with a second-order Israel–Stewart or BDNK form, and verify that the resulting system is causal and stable in the weak-field limit.
- **Write the one-speed law.** Write the Ocean's law out in full on its own field of rulers and clocks, with no fixed speed anywhere, and derive how the Ocean's state sets those rulers and clocks (Sections 11.1 and 11.4). The principle and its consequences for stiffness are in Section 11.1; the geometry around a non-spinning mass follows from two principles (Section 11.7); the general law does not yet. Volovik's helium-3 work is the standard example of a medium whose excitations share one emergent speed [16].
- **Establish whether matter coupling constitutes a forcing term.** The known finite-time blowup constructions for viscous media require an external force injecting momentum; the unforced problem remains open. In this framework the candidate forcing is the matter sector acting on the vacuum through $\kappa(\phi)T^{\text{matter}}_{\mu\nu}$ in (1). Determine whether that coupling can drive the medium to finite-time concentration, or whether the healing-length cutoff bounds it for any physically admissible matter distribution. This bears on the regularity of the medium. Whether a contraction phase turns around is a separate question: stiffness cannot do it, and the candidate is the loosening grip of Section 11.8; derive it, and show that it acts at the end of a contraction as at the core of a black hole.
- **Derive the viscosities from vortex dynamics.** Determine whether η and ζ follow from the creation, travel and reconnection of quantized vortices, as effective dissipation does in laboratory superfluids, removing two free functions from the theory.
- **Simulate a galactic disturbance.** Once the equation is fixed, solve the condensate wave equation (Gross–Pitaevskii or nonlinear Klein–Gordon type) for a galaxy-scale disturbance, including shear, wakes and vortex structure.
- **Reproduce the measured vacuum.** Its strengths would also fix how stiffly the Ocean resists a dent, separately from how much Ocean there is. Compute the Ocean's contribution to vacuum polarization, and hence to the Lamb shift and the anomalous magnetic moment, together with light-by-light scattering, vacuum birefringence and the Casimir force, and compare with quantum electrodynamics and with measurement (deficiency (4) of Section 10).
- **Size and spin of the localized excitation.** Compute the probe response of the localized excitation and show it is pointlike to about $10^{-18}\,\mathrm{m}$; determine how $\xi(\phi)$ behaves at high concentration consistently with the collapse floor; and identify the structure that supplies spin $\frac{1}{2}$.
- **Travel through the surroundings.** Determine the flow regimes of a disturbance traveling relative to the medium, including the superfluid critical speed. Travel through the Ocean below that speed is frictionless and leaves no trace; the microwave-background dipole shows the solar system traveling at about $370\,\mathrm{km\,s^{-1}}$ and the Milky Way at about $600\,\mathrm{km\,s^{-1}}$ relative to the frame in which that background is uniform, the natural candidate for the Ocean's local rest frame, and precision tests see nothing at those speeds, so the framework must produce a critical speed well above them. Test for residual wake offsets relative to the direction of travel; for residuals of the radial acceleration relation that track speed through the environment, a property not among those tested in [9]; and against colliding clusters.
- **Derive gravity from the medium.** Derive the effective geometry of Equation (1) from the dynamics of the medium, and show that the refractive account of Section 2.5 meets its three decisive requirements: every excitation responds in exactly the same proportion; time is slowed and space stretched in equal measure, giving the observed deflection of starlight; and gravitational waves travel at the speed of light. Then extend it to perihelion advance, frame dragging and gravitational radiation, or identify where it fails. Produce G with its observed value, equivalently α_G of (23), or identify the structure that sets it (Section 7.7, item (5)). Derive the principle of

Section 5.1: a gradient of state refracts without net flow, and a uniform change of state produces divergent or convergent flow. Derive Equation (1) with the background defining the metric and only excitations as sources (Section 11.3), under the one-speed law, including how $P(\phi)$ enters c_s and the conservation conditions that follow.

- **Derive the candidate mechanism's inputs.** The current through time (Sections 11.6 and 11.7) now needs two inputs: the first-order lockstep constant, equivalent to $\gamma = 1$, and the strength of the drag, which sets G. Derive both under the one-speed law; meet the bounds $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ and $\beta - 1 = (-4.1 \pm 7.8) \times 10^{-5}$ [52]; and show that the Ocean's second state variable keeps its cosmological pressure from being a function of density alone, as the exclusion of single-density unified fluids requires [56].

- **The strength of gravity from the drag.** If the candidate mechanism holds, G is set by the drag a pattern exerts on the current through time relative to the Ocean's viscosity. Determine whether that gives the observed α_G without structure near the Planck length.

- **Frame dragging.** Extend the candidate mechanism to a rotating source and show that it produces the gravitomagnetic effect measured by laser-ranged satellites and Gravity Probe B [52]; the nearest precedent lacks it [35].

- **One speed and the normal component.** A normal component, required for mutual friction, brings a second sound with a speed of its own [48]. Show that it carries no signal patterns can use, or that it shares c_s, or revise the coarse-grained account of $\Pi_{\mu\nu}$ (Section 2.3).

- **Gravity from the electron and the pressure.** Read the other way, the relation of Section 10.3 gives the strength of gravity from the electron mass and the cosmological constant, $G = \alpha^3 \hbar^2 \sqrt{\Lambda}/m_e^3$, which with the measured Λ comes within about ten per cent of the measured G, and within a few per cent if the locally measured expansion rate is used. Here m_e is set by the amount ρ and held for the beat (Section 7.6), while Λ is the present value of $P(\phi)$, which takes the cosmological constant's place (Section 2.4) and may evolve within the beat (Section 5.1). Determine whether the medium produces this relation, which would derive G rather than fit it. If it does, G follows the pressure, and the lunar-ranging bound of Section 7.6 limits how fast the pressure can change today; if not, the relation is a numerical coincidence of the present epoch. Trace it to the source that first derived it.

- **Black holes.** The geometric and medium horizons coincide (Section 8.1); determine what horizon-scale imaging and ringdown measurements constrain beyond the shadow. Explain why a black hole's capacity for state scales with horizon area. Compute the infall asymmetry and drag for a black hole traveling relative to the medium, including the critical-speed regime, and compare with high-velocity merger remnants. For a non-spinning mass the Ocean's geometry is General Relativity's, with the point of no return at $2GM/c_0^2$ and the observed shadow size (Section 11.7); extend it to spinning black holes, which needs frame dragging, and confront the ringdowns of mergers. Calculate the turnaround at the core and the time a black hole takes to return what it holds, which depend on what the Ocean is made of (Section 11.8). Revise the stages of Section 8.2 as these results arrive.

- **Antimatter and circulation sense.** From the complex-field formulation, derive charge-conjugate partners of identical mass and magnetic moment and opposite charge, and confront the predicted CPT and Lorentz violation with the limits of Section 10, deficiency (5). Identify the structure that makes circulation sense invariant under rotation, the conserved quantity that passes between localized and coherent excitations (Section 7.7, items (8)–(9)), and the long-range gauge field that electric charge requires.

- **Order parameter.** Determine whether a multi-component order parameter can supply spin $\frac{1}{2}$, a second internal phase and protected relativistic dispersion, and what it costs the framework's

foundations (Section 7.7, item (7)). Light's transverse polarizations and the tensor polarizations of gravitational waves make the same demand (Section 11.9).

- **Viscosity as scrambling.** Show that $\Pi_{\mu\nu}$ arises as the coarse-grained description of reversible dynamics (Section 2.3), together with the item on viscosities from vortex dynamics, and define the normal component, the excitation gas distinct from the condensate, that mutual friction requires.
- **The redshift budget.** Show where the energy lost by radiation crossing an expanding region goes, consistently with preservation, and confront any exchange term with the microwave background and light-element abundances (Section 9.3).
- **Galactic shear and coupling.** Compute the anisotropic $\Pi_{\mu\nu}$ of a flattened vortex lattice bound to a disc galaxy and, in the geometry patterns read (Section 11.3), the radial-to-vertical ratio of the acceleration it produces, and confront that ratio with Milky Way kinematics; and specify how the travel of stars and gas couples to the flow of the medium (Section 6). For the column-like swirl of Section 6, test the dependence of the extra pull on how a star travels through the swirl against Gaia's velocity data, check the asymmetric deflection of light against galaxy lensing, and find what sets the swirl's strength from a galaxy's baryons.
- **Flow on the largest scales.** A smooth flow is invisible; only shear, gradients and eddies can be seen. Formulate source-free currents of the Ocean, driven by differences of chemical potential and shaped by quantized circulation, and determine what a beat transition leaves of them. Within the solar system they are bounded by preferred-frame tests; on large scales they would appear as coherent streams of galaxies that the mass distribution does not account for. Compare the measured large-scale flow of galaxies with the flow predicted from their mass, and test for coherent rotation beyond what the mass predicts.
- **Contraction-phase contact and the age of the Ocean.** Show that contact during the previous contraction produces the uniformity and fluctuation spectrum of the microwave background, including the baryon acoustic scale; and determine whether disorder accumulates from beat to beat and whether the Heartbeat has net expansion, on which the Ocean's having an age depends (Section 5.2).
- **The heartbeat mechanism.** Model the exhaustion of the gradients that drive expansion, the residual self-attraction of the medium and a stiff contraction, and determine the conditions under which contraction stays smooth. Determine what sets μ today, whether the Ocean's rest frame is the frame in which the microwave background is uniform, and whether the fall of μ as the Ocean expands is what ends expansion, with μ as the clock of a beat and ρ as what resets at the transition (Section 2.2).
- **The cost of a dent.** In the stiff Ocean the gap of the amplitude mode of ρ is $\sqrt{\lambda}\,v$ and the healing length is its inverse, $\xi = \hbar c_0/E_{\text{dent}}$: a dent costing 0.511 MeV gives 386 femtometers. So the healing length rests on that gap, not on the k^4 term; but matching 0.511 MeV fixes only the combination $\sqrt{\lambda}\,v$, a calibration, not a derivation. Stiffness also bounds the Ocean's turning rate: $\mu \lesssim 10^{-6}\,\text{eV}$. Fix λ or v separately, for instance from the measured vacuum above.
- **The matter excess across a beat.** Compute the balance between erasure of the matter excess by black holes, which strip lepton number, and its survival in excitations that escape capture and annihilation.
- **Hearing the medium.** If the gapless phonon of the phase (Section 2.2) couples to matter, it implies a weak long-range force or an energy-loss channel. Compare with torsion-balance and atom-interferometer bounds.

13 Summary of the Revised Formalism

The revised Ocean framework rests on a single premise: there is no empty space. The Ocean is one continuous medium without edges or outside, and everything in it, matter and light included, is a pattern of that medium. Gravity is proposed to be the medium's response to concentrated energy. The mathematical formulation keeps Einstein's equation as the effective description of that response, with a scalar-field stress sector, a viscous constitutive sector, a variable pressure contribution and a field-dependent effective coupling as the medium's contributions to it.

The framework extends that idea to the matter sector itself. If the vacuum is a medium, it has a characteristic length over which it can heal; excitations shorter than that length are localized, and excitations longer than it are coherent excitations of the medium. The distinction between matter and vacuum then becomes a statement about the medium's own excitation spectrum rather than an imposed energy cut.

Three principles frame the account. The Ocean has one speed, at which every disturbance travels, and gravity is where that speed varies from place to place; no speed is fixed. One change of state, in two shapes, makes both gravity and expansion: a gradient of state refracts, and a uniform change of state flows. And the state of the Ocean is never destroyed, only its patterns scrambled; black holes recycle energy and keep no structure.

The central hypothesis is that phenomena conventionally divided into dark energy, dark matter, and localized matter may have a common origin in different dynamical regimes of one vacuum field. Establishing that claim requires deriving the relevant limits and demonstrating quantitative agreement with observations.

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Reference basis. Revised from an earlier formalism supplied by the author, which defined the Ocean metric, the vacuum viscosity tensor, the scalar pressure term, the effective coupling $\kappa(\phi)$, a variable-speed-of-light reading of $c(\phi)$, and a 0.511 MeV matter threshold. Two of these are superseded: the threshold by Section 7, and the variable-speed-of-light reading by Section 11.1, where $c(\phi)$ is the Ocean's one carrying speed rather than a second causal speed.